

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Adaptive Weight-Only Quantization for QKV Projections in Grouped Query Attention via Flip/ReFlip

LƯU THIÊN VIỆT CƯỜNG

cuong.ltv225477@sis.hust.edu.vn

Program: DS&AI

Supervisor: Dr. Ngô Văn Linh

Department: DS&AI

School: School of Information and Communications Technology

HANOI, 03/2026

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

**Adaptive Weight-Only Quantization for QKV
Projections in Grouped Query Attention via
Flip/ReFlip**

LƯU THIÊN VIỆT CƯỜNG

cuong.ltv225477@sis.hust.edu.vn

Program: DS&AI

Supervisor: Dr. Ngô Văn Linh _____

Signature

Department: DS&AI

School: School of Information and Communications Technology

HANOI, 03/2026

ACKNOWLEDGMENT

I would like to express my deepest and most sincere gratitude to my supervisor, Dr. Ngô Văn Linh, for his invaluable guidance, thoughtful feedback, and continuous support throughout the entire process of completing this thesis; his dedication, patience, and academic rigor have not only helped shape the direction and quality of this work but have also greatly influenced my way of thinking and approaching research problems. I would also like to extend my heartfelt appreciation to all the lecturers and faculty members whom I have had the privilege to learn from during my time at the School of Information and Communications Technology, Hanoi University of Science and Technology; their knowledge, passion for teaching, and commitment to academic excellence have built a strong foundation for my development, and I truly hope to have more opportunities in the future to continue learning from them and, if possible, to collaborate and work alongside them in a professional environment. I am deeply grateful to my parents for their unconditional love, unwavering support, and constant encouragement throughout my academic journey. I also sincerely thank my friends for their companionship, support, and shared experiences, which have made this journey both meaningful and memorable.

ABSTRACT

Large Language Models (LLMs) require substantial parameter memory and memory bandwidth during inference, which limits their deployment under constrained hardware and high-throughput serving conditions. Post-training weight-only quantization reduces these costs without full model retraining, but aggressive low-bit settings can introduce errors whose effects are not fully described by the isolated reconstruction error of each weight. In attention projections, signed weight errors interact with activations and may accumulate or cancel in the projected query, key, and value representations. Grouped Query Attention (GQA) adds further structure because several query heads share one key-value head, making the roles of the three projections distinct.

This thesis studies post-training, group-wise asymmetric INT4 weight-only quantization for QKV projections and presents two separate discrete correction stages. Flip selectively changes round-to-nearest decisions when the change reduces an aggregate activation-weighted projection residual, even if the isolated error of an individual weight increases. ReFlip is a subsequent query-only correction. For each query head in a GQA group, it holds the shared quantized key projection fixed and evaluates candidate changes to the quantized query projection through an unscaled scalar Q–K score surrogate. Kneedle-based sensitivity partitioning is used to identify a moderate candidate region for ReFlip; this partition narrows the search but does not provide an optimality guarantee. Statistical shrinkage, including the motivation of James–Stein estimation, informs the construction of a stable representative activation but is not used to claim a classical estimation guarantee for either correction stage.

The experimental evaluation separates local score-level evidence from model-level behavior. A standalone GQA case study compares round-to-nearest quantization, Flip, and Flip followed by ReFlip under a controlled INT4 setting, while full-model comparisons examine perplexity and selected downstream task metrics across the evaluated model and quantization configurations. The reported results provide evidence that the two correction objectives can reduce their targeted forms of quantization distortion in the evaluated settings, while the magnitude and consistency of model-level changes remain dependent on the model, baseline quantizer, and task. These findings support targeted post-quantization correction as a useful direction while also establishing clear limits: the scalar ReFlip objective is a surrogate rather than the complete attention operation, Kneedle performs candidate

selection rather than global optimization, and local improvements do not guarantee universal downstream gains.

Student
(Signature and full name)

TABLE OF CONTENTS

CHAPTER 1. INTRODUCTION.....	1
1.1 Problem Statement.....	1
1.2 Background and Research Motivation.....	2
1.3 Research Questions and Objectives	4
1.4 Scope and Boundaries.....	5
1.5 Contributions	5
1.6 Organization of the Thesis	6
CHAPTER 2. LITERATURE REVIEW	7
2.1 Scope of Research	7
2.2 Quantization Concepts and Taxonomy	9
2.3 Related Work and Evidence Boundaries.....	11
2.4 Group-wise Asymmetric Weight-only Quantization	14
2.5 Attention Computation and Projection Perturbations.....	16
2.6 GQA and Dependent Error	19
2.7 Outliers, Saliency, and Candidate Selection.....	20
2.8 Selective Adjacent-level Refinement.....	22
2.9 Statistical Interpretation and Its Limits	23
2.10 Synthesis and Research Gap	24
CHAPTER 3. METHODOLOGY	27
3.1 Overview	27
3.2 Notation and Calibration Representative	28
3.3 Problem Formulation	30
3.4 Group-Wise Asymmetric INT4 Quantization	32
3.5 Flip: Activation-Weighted Projection Refinement	34

3.6 ReFlip: Scalar Query–Key Surrogate Refinement	37
3.7 Unified Refinement Procedure	42
3.8 Hyperparameters and Design Choices	44
3.9 Theoretical Discussion	45
3.10 Summary	47
CHAPTER 4. EXPERIMENTAL EVALUATION.....	50
4.1 Experimental Setup.....	50
4.2 Standalone Q–K Surrogate Evaluation.....	52
4.3 Full-Model Evaluation	55
4.4 Discussion	57
CHAPTER 5. CONCLUSIONS	61
5.1 Summary	61
5.2 Main Findings	62
5.3 Limitations.....	63
5.4 Future Work.....	64
REFERENCE	67

LIST OF FIGURES

Figure 4.1	Standalone Q–K surrogate error comparison	54
Figure 4.2	Q–K surrogate correction behavior	54

LIST OF TABLES

Table 4.1	Standalone case-study configuration	51
Table 4.2	Standalone scalar Q–K surrogate error	53
Table 4.3	Per-head scalar Q–K surrogate errors	53
Table 4.4	4-bit evaluation for Mistral-7B	56
Table 4.5	4-bit evaluation for Llama-3.1-8B	56
Table 4.6	4-bit evaluation for Qwen2.5-7B	57
Table 4.7	4-bit evaluation for Llama-3-8B	57

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Large Language Models (LLMs) have achieved strong performance across a broad range of natural language processing tasks. Their practical deployment, however, remains constrained by the memory capacity, memory bandwidth, and computational resources required during inference. These constraints are especially important when a model must operate on limited hardware, serve many concurrent requests, or satisfy strict latency and energy budgets. Reducing the storage and data-movement cost of model parameters is therefore an important direction for making LLM inference more accessible and efficient.

Post-training quantization (PTQ) reduces these costs by representing pretrained model parameters with lower-precision numerical formats after training, without requiring full model retraining. Weight-only PTQ is particularly attractive because it compresses the stored model weights while retaining higher-precision activations, thereby reducing parameter memory and data movement while limiting changes to the surrounding inference computation [1], [2], [3]. Nevertheless, aggressive low-bit quantization can introduce substantial approximation error. At INT4 precision, the errors contributed by many weights may accumulate under the activations observed during inference, and the resulting aggregate projection residual can vary considerably across model components even when every individual weight has been rounded to its nearest representable level.

This thesis studies group-wise asymmetric INT4 weight-only quantization of the query, key, and value projection matrices in Transformer attention. In this setting, each weight matrix is partitioned into groups, and each group is represented by its own scale and asymmetric zero-point. The group-wise formulation can adapt to local weight distributions more effectively than a single matrix-wide quantizer, but it does not eliminate quantization error. Attention projections are a consequential target because perturbations to queries and keys alter attention scores, while perturbations to values alter the information aggregated by attention.

The notation used throughout this thesis treats a representative activation token x as a row vector. The corresponding projected row vectors are

$$q = xW_Q^T, \quad k = xW_K^T, \quad v = xW_V^T.$$

For a projection matrix W and its quantized counterpart $\widehat{W}$, the activation-weighted

projection residual for a token is $x(\widehat{W} - W)^T$. Each output coordinate of this residual aggregates the signed contributions of many individual weight errors after they have been weighted by the corresponding activation values. Consequently, the isolated error of each quantized scalar is not, by itself, a complete measure of the functional projection error. Changes to W_Q additionally change the query representation and the scores formed with keys. These observations motivate two targeted but distinct correction procedures. Flip reduces an aggregate activation-weighted projection residual. ReFlip subsequently updates only the quantized query projection and evaluates each candidate through an unscaled scalar Q–K score surrogate for a representative activation. The central problem is therefore to determine whether lightweight, structure-aware corrections can improve the fidelity of group-wise asymmetric INT4 QKV quantization without expanding the correction objectives into full model retraining or claiming to optimize the complete attention operation.

1.2 Background and Research Motivation

The Transformer architecture relies on attention to determine how token representations exchange information. Standard multi-head attention assigns separate query, key, and value heads. Multi-query attention and Grouped Query Attention (GQA) reduce key-value redundancy by allowing multiple query heads to share a key-value head [4], [5]. A GQA group therefore contains several query heads associated with one shared key head and one shared value head. This structure can reduce key-value storage and improve decoding efficiency, but it also creates an asymmetric error relationship: a perturbation in a shared key or value representation may influence several query heads, whereas a perturbation in one query head remains local to that query head. Quantization methods applied to GQA should therefore account for the distinct roles and shapes of the QKV projections rather than treating them as interchangeable matrices.

Low-bit quantization must balance representational range against resolution. Asymmetric quantization can better represent groups whose distributions are not centered at zero, while group-wise quantization limits the effect of distribution differences across a large matrix. A standard round-to-nearest (RTN) decision selects the nearest representable quantization level for each weight and therefore minimizes the isolated scalar quantization error for that weight under the fixed quantizer. RTN does not, however, directly minimize the aggregate activation-weighted residual because signed scalar errors can reinforce or cancel after multiplication by the calibration activations. Static criteria for identifying consequential rounding decisions are not always appropriate because activation patterns and residual contributions can differ between matrices, layers, and groups. This thesis conse-

quently considers adaptive candidate-region identification based on the Kneedle algorithm [6], which identifies a knee in an ordered sensitivity curve. For ReFlip, the detected knee provides a data-dependent boundary between extremely sensitive query dimensions that are protected and a moderate post-knee region considered for correction. This partition narrows the search; it neither proves that the selected region contains the globally best correction nor gives an optimality guarantee for the resulting discrete updates.

The proposed framework distinguishes two correction stages. *Flip* changes selected RTN rounding decisions after group-wise asymmetric INT4 quantization in order to reduce the aggregate activation-weighted projection residual under a representative activation constructed from calibration data. Because RTN already minimizes each isolated scalar error, a flipped decision may move an individual quantized weight farther from its original full-precision value. Flip accepts such an increase only when the changed signed contribution helps reduce the combined residual produced by the relevant weights under that representative activation. It therefore targets cancellation and accumulation in the aggregate residual rather than seeking uniformly smaller individual weight errors. Flip preserves the INT4 representation and the broader quantization configuration, and it does not use the attention-score objective described below.

ReFlip is a separate, subsequent correction stage. ReFlip operates per query head within a GQA group and updates only the corresponding query projection W_Q ; it does not update the shared W_K , the shared W_V , or the other query heads while evaluating that head’s correction. The update is guided by a scalar score loss that measures disagreement between reference and quantized Q–K dot products for a representative activation. This unscaled scalar surrogate isolates a specific bilinear effect of query-projection error, but it is not the complete attention-logit matrix, the softmax attention distribution, or the final attention output. The method is also informed by the general motivation of statistical shrinkage: in high-dimensional estimation, carefully regularized adjustments can, under appropriate assumptions, reduce aggregate error relative to unregularized estimates [7]. This connection motivates the construction of a stable representative activation; it is not a claim that Flip or ReFlip is a James–Stein estimator or inherits a classical risk guarantee.

Together, Flip and ReFlip address different aspects of the same problem. Flip changes selected rounding decisions across QKV projections to reduce an aggregate activation-weighted projection residual, whereas ReFlip focuses narrowly on one query head at a time and uses scalar Q–K score disagreement as its signal. A candidate that benefits ReFlip may therefore fail to reduce the projection residual

targeted by Flip, just as a candidate that benefits Flip need not reduce the score-level surrogate. Keeping the two stages conceptually separate is important for evaluating their individual effects and for avoiding claims that improvements under one objective necessarily imply improvements under the other or under end-to-end evaluation.

1.3 Research Questions and Objectives

The overall objective of this thesis is to investigate whether targeted corrections can improve the fidelity of group-wise asymmetric INT4 weight-only quantization for QKV projections, particularly in attention modules using GQA. The study is guided by the following research questions:

1. Why can RTN’s isolated scalar-error objective leave an aggregate activation-weighted projection residual or scalar Q–K score error after group-wise asymmetric INT4 weight-only quantization?
2. Can selectively accepted Flip decisions reduce aggregate activation-weighted residuals for selected attention projections—specifically the Q and K projections examined in the standalone study—even when some changed decisions increase their individual scalar weight errors, without changing the prescribed INT4 weight-only setting?
3. Can query-only ReFlip corrections, evaluated per query head through a scalar Q–K score surrogate, reduce the score error remaining after Flip?
4. What patterns are observed when RTN, Flip, and Flip followed by ReFlip are compared in the standalone score-level case study and in the reported full-model evaluations?

To answer these questions, the thesis develops a controlled framework that first establishes the INT4 quantization setting, then evaluates Flip and ReFlip as distinct interventions. The analysis distinguishes isolated scalar quantization error, aggregate activation-weighted projection residual, scalar score-surrogate error, and downstream evaluation. This distinction is necessary because RTN can minimize each isolated scalar error while leaving a larger combined projection residual, because a reduction in that residual does not by itself guarantee a reduction in score error, and because a local score reduction does not by itself guarantee improved model-level behavior. The reported comparisons are consequently interpreted as empirical evidence within the stated settings rather than as universal performance guarantees.

1.4 Scope and Boundaries

This work is limited to post-training, group-wise asymmetric INT4 weight-only quantization. Activations remain outside the low-bit quantization scope, and the thesis does not propose quantization-aware training or full model retraining. The primary objects of study are the W_Q , W_K , and W_V projection matrices in Transformer attention, with particular attention to GQA structure. A quantization group refers to the subset of weights sharing quantization parameters, whereas a GQA group refers to multiple query heads sharing one key-value head; these two forms of grouping serve different purposes. Other model components may still affect end-to-end behavior, but they are not the central correction targets considered here.

Flip and ReFlip are deliberately bounded procedures. Flip changes only selected RTN rounding decisions within the quantized QKV projections and selects those changes according to their effect on the aggregate activation-weighted projection residual under a representative activation constructed from calibration data. It may worsen individual scalar weight errors and is not intended to minimize a weight-wise error metric. ReFlip updates only W_Q , one query head at a time, and selects accepted updates using an unscaled scalar Q–K score surrogate under the same representative activation; it does not update W_K , W_V , or the remaining model parameters. The surrogate does not represent the full sequence-to-sequence attention matrix, softmax normalization, or value aggregation. Kneedle identifies a moderate candidate region rather than an optimal solution, and the subsequent accepted updates remain a local discrete correction. The thesis evaluates these procedures empirically and does not claim universal improvement across all architectures, tasks, datasets, group sizes, activation distributions, or calibration choices. Likewise, adaptive selection, representative-activation construction, and shrinkage-inspired reasoning are treated as practical design principles whose usefulness must be established by the reported experiments.

1.5 Contributions

Within the stated scope, this thesis makes the following contributions:

1. It formulates a targeted study of QKV projections under group-wise asymmetric INT4 weight-only quantization, with explicit consideration of the sharing structure present in GQA.
2. It defines Flip as a selective change to RTN rounding decisions that may increase individual scalar weight errors while reducing an aggregate activation-weighted projection residual, and it evaluates this objective on the selected Q and K projections considered in the standalone study while distinguishing it

from ReFlip.

3. It defines ReFlip as a per-query-head correction that updates W_Q using an unscaled scalar Q–K score surrogate while leaving the shared W_K , shared W_V , and other model parameters unchanged.
4. It incorporates Kneedle-based partitioning of query-dimension sensitivities so that ReFlip protects the extreme pre-knee region and evaluates candidates from a moderate post-knee region, while explicitly treating this partition as candidate selection rather than an optimality guarantee.
5. It provides an empirical assessment at both a standalone score-surrogate level and a reported full-model level, explicitly distinguishing RTN’s isolated scalar-error objective, Flip’s aggregate activation-weighted residual objective, and ReFlip’s scalar score objective rather than assuming that improvement under any one objective necessarily produces better end-to-end performance.

These contributions are intended as a focused investigation of attention-projection correction under a specific low-bit setting. They do not establish that QKV projections are always the dominant source of quantization degradation or that the proposed procedures replace broader post-training quantization methods.

1.6 Organization of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 reviews Transformer attention, GQA, and relevant approaches to neural network and LLM quantization. Chapter 3 presents the proposed methodology, including group-wise asymmetric INT4 weight-only quantization, the aggregate residual objective of Flip, sensitivity-based candidate selection, and the per-query-head scalar Q–K surrogate used by ReFlip. Chapter 4 reports the experimental evaluation through a standalone GQA score-level case study and full-model comparisons based on perplexity and selected downstream tasks, while discussing the boundaries of the evidence. Chapter 5 summarizes the findings and limitations and identifies directions for future research.

CHAPTER 2. LITERATURE REVIEW

This chapter establishes the conceptual and empirical context for low-bit quantization of attention projection weights in large language models. The discussion first defines the scope of the thesis and then distinguishes the principal quantization taxonomies that are often conflated in the literature. It subsequently reviews representative post-training quantization methods, develops the mathematical background of attention and Grouped Query Attention (GQA), and analyzes how projection-weight perturbations enter attention scores and outputs. The final sections separate weight outliers, activation outliers, and functional salience; clarify the limited roles of Kneedle and James–Stein reasoning; and identify the research gap addressed by the thesis. Throughout the chapter, claims about sensitivity and error propagation are qualified by the relevant operating conditions. This distinction is important because structural reuse creates dependencies, but does not by itself prove that errors will be amplified in every input, head, layer, or task.

2.1 Scope of Research

This thesis studies post-training quantization of the query, key, and value projection weights in Transformer-based large language models, with particular attention to models that use GQA. The target setting is group-wise asymmetric INT4 weight-only quantization. The original pretrained model is not retrained or fine-tuned. Activations remain in higher precision, and the study does not attempt to optimize every layer or every inference-system component. This restricted scope makes it possible to examine how the statistical structure of QKV weights and the computational structure of attention jointly affect quantization error.

The models motivating the study include contemporary Transformer families such as LLaMA [8] and Qwen [9]. Their parameter counts make memory capacity and memory bandwidth important deployment constraints. Weight-only quantization directly reduces the storage and transfer cost of model parameters while avoiding the additional dynamic-range difficulties introduced by quantizing activations [1], [2], [3], [10]. The resulting inference process still uses higher-precision token representations, but each linear projection is formed from an approximation of its original weight matrix. Consequently, the quality of the projection depends on both the quantized weights and the activations that multiply them.

The thesis quantizer is consistently defined as *group-wise asymmetric weight-only quantization*. For a projection matrix, each row corresponds to an output channel and each column corresponds to an input channel. Contiguous weights along

the input-channel dimension are partitioned into numerical quantization groups. Every such group receives its own scale and zero-point. These numerical groups are unrelated to GQA groups. A GQA group is an architectural association in which several query heads share a key head and a value head; a quantization group is merely a local partition of weights represented by common affine parameters. Maintaining this distinction prevents ambiguity when discussing group size, error sharing, or head structure.

The study also considers a lightweight post-quantization refinement referred to as *bit-level adjustment*. In this thesis, the term has a precise operational meaning: a selected stored integer weight value is changed to an adjacent integer code, and therefore to an adjacent representable quantization level, while retaining the group’s scale, zero-point, and assigned bit width. Bit-level adjustment is not a literal flip of a particular binary digit. A literal binary-digit flip can move an integer by a large and representation-dependent distance, whereas the intended operation changes the quantized value by one neighboring level. The term is preserved because it emphasizes discrete low-bit refinement, but every subsequent use refers to adjacent integer-code adjustment.

Calibration data may be used to characterize the effects of quantized weights under representative inputs and to evaluate candidate adjustments. Such use does not constitute training: no gradient-based optimization of the original model parameters is required. The study also excludes hardware-specific kernel design and activation or KV-cache quantization. These topics are relevant to complete deployment systems, but separating them from projection-weight quantization keeps the evidence boundaries clear. In particular, results about quantized KV caches can motivate attention-aware analysis, yet they do not directly establish how QKV projection weights behave under group-wise asymmetric weight-only quantization.

The resulting research question is deliberately narrow. Given low-bit group-wise asymmetric representations of QKV projection weights, can selective adjacent-level adjustment reduce consequential residual errors while respecting the structure of attention and GQA? Answering this question requires more than reporting weight reconstruction error. It requires a taxonomy of quantization choices, a perturbation model that connects weight error to attention computations, and careful interpretation of statistical heuristics. These foundations are developed in the remainder of the chapter.

2.2 Quantization Concepts and Taxonomy

Quantization maps values from a high-precision domain to a finite representable set. For neural-network weights, this mapping reduces memory consumption and can reduce data-transfer cost, but it introduces approximation error. The literature classifies quantizers along several independent dimensions, including uniform versus non-uniform level spacing, symmetric versus asymmetric ranges, tensor granularity, weight-only versus weight-and-activation quantization, and post-training versus training-aware procedures [1], [2]. A precise account must keep these dimensions separate because a quantizer can, for example, be uniform, asymmetric, group-wise, weight-only, and post-training at the same time.

Uniform and Non-uniform Levels

A uniform quantizer uses a constant step size between adjacent dequantized values. If the step size is $s > 0$, moving from one integer code to its neighbor changes the dequantized value by exactly s , except where clipping prevents movement beyond the representable endpoints. Uniformity refers only to spacing. It does not require the range to be centered at zero, nor does it require all weights in a tensor to share one scale.

A non-uniform quantizer places representable values at unequal intervals. Such placement can allocate more levels to dense regions of a distribution and fewer levels to sparse regions, potentially improving representation for a specified distortion measure. The trade-off is that irregular levels can complicate representation and arithmetic. The thesis does not use a non-uniform quantizer. Its asymmetric affine quantization remains uniformly spaced within every quantization group.

Symmetric and Asymmetric Affine Quantization

Symmetry concerns the location of the represented range relative to zero. A symmetric quantizer ordinarily uses a range centered at or approximately centered at zero. This choice can simplify some arithmetic and is appropriate when positive and negative magnitudes are balanced. An asymmetric affine quantizer introduces a zero-point that shifts the represented interval. The shift can use the available integer range more effectively when a local weight group is skewed or has unequal positive and negative extrema.

Let a quantization group contain real-valued weights w_i , and let the available integer interval be $[c_{\min}, c_{\max}]$. Following the standard integer-arithmetic formulation [11], affine quantization and dequantization with scale $s > 0$ and integer zero-point

$z \in \mathbb{Z}$ can be written as

$$c_i = \text{clip} \left(\text{round} \left(\frac{w_i}{s} \right) + z, c_{\min}, c_{\max} \right), \quad \hat{w}_i = s(c_i - z). \quad (2.1)$$

The representable dequantized values are separated by the constant distance s . Thus, asymmetric affine quantization is still uniform quantization. Its asymmetry shifts the represented range through z ; it does not create unequal spacing. For INT4, at most $2^4 = 16$ integer codes are available per group, so the choice of range and the treatment of extreme weights strongly influence the resolution assigned to the remaining weights.

The quantization error for an individual weight is $\Delta w_i = \hat{w}_i - w_i$. Before clipping, nearest rounding bounds its magnitude by approximately $s/2$. Clipping can produce a larger error because a value outside the selected range is mapped to an endpoint. This elementary distinction is important when interpreting candidate refinement: adjacent-level adjustment changes $\hat{w}_i$ by s , whereas changing the scale or clipping range changes the representable set for every weight in the group.

Granularity: Tensor, Output Channel, and Quantization Group

Quantization granularity identifies which weights share affine parameters. Per-tensor quantization assigns one scale and, where applicable, one zero-point to an entire tensor. It is economical but forces heterogeneous regions to share one range. Per-channel quantization, as a taxonomy comparator, assigns separate parameters to complete channels. It can adapt to differences across channels but still treats every weight within one channel as a single range.

The thesis instead uses group-wise quantization. Within each output-channel row, contiguous columns along the input-channel dimension are divided into quantization groups. Each group has an independent scale and zero-point. This arrangement provides finer local adaptation than per-tensor or complete-channel granularity while preserving a regular partition. The phrase “group-wise” always refers to this numerical partition unless “GQA group” is explicitly stated.

The distinction between input and output channels follows the orientation of the projection matrices used throughout this chapter. For one token representation $x \in \mathbb{R}^{d_{\text{in}}}$ and a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the projection is $xW^\top$. A row of W produces one output-channel component; a column multiplies one input feature. Grouping along the input-channel dimension therefore divides each row into contiguous sets of contributing input features. It does not combine multiple attention heads merely because those heads belong to the same GQA group.

Weight-only and Post-training Quantization

Weight-only quantization stores weights at reduced precision while leaving run-time activations in higher precision. The approximation error arises from the quantized weights, but its realized effect depends on the activations that multiply those weights. For a linear projection $y = xW^\top$ and quantized weights $\hat{W} = W + \Delta W$,

$$\hat{y} - y = x\Delta W^\top. \quad (2.2)$$

The same weight perturbation can therefore have different output effects for different inputs. A large-magnitude weight error aligned with rarely active input directions may have limited realized influence, whereas a smaller error aligned with frequently large or task-relevant activations may matter more. This relationship motivates the use of representative calibration inputs without changing the definition of the method as weight-only quantization.

Post-training quantization operates on a pretrained model without repeating the original training process. It can use weight statistics, calibration activations, local reconstruction objectives, or approximate second-order information. Quantization-aware training instead exposes model optimization to quantization effects and updates parameters accordingly. The post-training setting is attractive for large language models because full retraining is expensive, but the absence of retraining also makes the initial quantizer and any discrete refinement particularly important.

A basic post-training baseline is round-to-nearest (RTN) [1], [2]. Once a quantization group’s representable grid, scale, zero-point, and clipping interval have been fixed, RTN assigns each scalar weight to a nearest valid grid point. For any scalar and a fixed valid grid, including a value mapped to a representable endpoint, this assignment minimizes that scalar’s absolute and squared reconstruction errors among the available levels. This statement is deliberately local: it does not imply that independent nearest assignments minimize an activation-weighted projection error, a coupled attention-score error, or a downstream task objective. Those broader objectives contain activation-dependent weighting, correlations, and interactions among multiple quantized parameters.

2.3 Related Work and Evidence Boundaries***General Post-training Weight Quantization***

Post-training quantization methods differ in how they define the representable grid, select ranges, account for activations, choose rounding assignments, and compensate for local error. RTN provides the simplest reference point: after the grid is

fixed, it independently selects a nearest representable value for each scalar weight. More adaptive methods deliberately move beyond this independent fixed-grid decision because a set of scalar-nearest assignments need not minimize the error of a layer output or a downstream computation.

AdaRound [12] makes this distinction explicit by learning adaptive rounding decisions under a local reconstruction objective instead of retaining deterministic nearest rounding for every weight. It demonstrates that the choice between neighboring quantization levels can be optimized jointly with respect to layer behavior rather than only individual scalar distance. AdaRound is therefore an important conceptual predecessor for discrete post-quantization refinement. Its learned soft-to-hard rounding procedure and layer-reconstruction objective nevertheless differ from a restricted adjacent-level correction evaluated under explicitly defined projection or query–key interaction objectives.

GPTQ [13] formulates one-shot weight quantization using approximate second-order information and sequential error compensation. It processes weights in blocks and uses calibration-informed curvature information to reduce the reconstruction impact of decisions as quantization proceeds. It is consequently more precise to describe GPTQ as an approximate-second-order, block-wise reconstruction method than as a method that treats an entire layer with one homogeneous scale or independently applies RTN without compensation. Its reconstruction objective is a strong general baseline, but it is not specifically formulated as a post hoc correction of a scalar query–key interaction surrogate within a GQA group.

AWQ [3] uses activation statistics to identify activation-salient weights or feature channels and applies equivalent scaling transformations intended to protect their contributions during weight-only quantization. Here, “channel” denotes a projection feature direction, not a numerical quantization group and not a GQA group. AWQ therefore does not ignore channel differences, and criticism of it must not imply that it uses only a global layer scale or lacks activation awareness. Its salience criterion and scaling strategy address a different problem from explicitly evaluating coupled query–key score perturbations or the consequences of shared K/V heads in GQA. The existence of this difference motivates investigation of attention-specific refinement, but does not establish that AWQ is generally inadequate.

SmoothQuant [10] addresses weight-and-activation quantization by applying a mathematically equivalent channel-wise rescaling that transfers quantization difficulty from activations to weights. It is especially relevant to the distinction between activation outliers and weight outliers: large activation features can be made easier

to quantize at the cost of altering the weight-side range before quantization. The present thesis leaves activations in higher precision and refines already quantized projection weights, so SmoothQuant provides conceptual context rather than a directly matched weight-only or post-rounding baseline.

These methods demonstrate that rounding objectives, reconstruction criteria, activation statistics, curvature approximations, and equivalent transformations can all inform low-precision representation. They also illustrate why broad statements that existing methods are merely layer-wise, statistically unaware, or limited to independent RTN are inaccurate. The narrower research gap considered by this thesis concerns two explicit correction mechanisms after initial group-wise asymmetric weight-only quantization: aggregate activation-weighted projection-residual correction and query-only correction under a scalar query–key interaction surrogate. The gap is not the general idea of adaptive rounding or activation-aware quantization. It is the use of these particular objectives, their restricted adjacent-level actions, and their interpretation within the shared K/V structure of GQA. Whether these choices are useful is an empirical question rather than an assumption.

Adjacent Evidence from KV-cache Quantization

KVQuant [14] and QAK [15] examine quantization associated with key and value cache representations. Their results are relevant because the KV cache participates directly in attention over previously processed tokens, and long contexts repeatedly reuse stored representations. However, KV-cache quantization and QKV projection-weight quantization intervene at different locations and quantize different mathematical objects. Cache quantization approximates token-dependent key and value representations after projection and stores those approximations across decoding steps. Projection-weight quantization approximates the persistent matrices W_Q , W_K , and W_V that generate new queries, keys, and values from each input representation. The former changes stored activations; the latter changes the linear maps applied whenever projections are formed.

The two settings share downstream attention equations, so KV-cache evidence supports the general proposition that perturbations to keys and values can matter. Their error structures nevertheless differ. A cached-representation error is tied to particular tokens and decoding states, whereas one projection-weight error can affect every newly produced representation whose activation has a component along the corresponding input channel. Conversely, cache reuse across many decoding steps creates a temporal persistence that is not equivalent to repeatedly applying a quantized projection matrix. KV-cache results therefore do not directly prove that

projection weights are more sensitive than other weights, that a specific QKV projection must receive special treatment, or that a cache-oriented technique transfers to group-wise asymmetric weight-only quantization. Such conclusions require direct analysis or evaluation in the projection-weight setting. Treating KVQuant and QAA as adjacent rather than direct evidence preserves their relevance without extending their claims beyond their experimental object.

Attention Architectures

Standard multi-head attention (MHA) gives each attention head its own query, key, and value head [16]. The separate heads can learn different relations and produce different attention distributions. This structure does not guarantee that perturbations remain isolated, because head outputs are later combined and transformed, but head-specific K/V representations do limit direct sharing at the score-computation stage.

Multi-query attention (MQA) shares one key head and one value head across many query heads [4]. GQA occupies an intermediate design point: several query heads are assigned to each shared key/value pair [5]. This reduces KV-cache size and can improve inference efficiency while retaining more K/V diversity than MQA. In GQA terminology, the query heads assigned to one shared key/value pair form a GQA group.

Sharing changes the dependency structure of perturbations. If a shared key is perturbed, every query head associated with that key computes scores using the same perturbed representation. Likewise, a shared value perturbation enters the weighted aggregation for every associated query head. The resulting errors are dependent because they contain a common component. Dependence does not imply that their magnitudes add monotonically or that the final model error is necessarily larger than in MHA. Query directions can make the shared key perturbation influential for some heads and nearly orthogonal for others; softmax can attenuate or redistribute score changes; value effects can cancel after output projection; and later layers can suppress or compound the perturbation. GQA therefore creates a pathway for conditional amplification and coordinated error, not a universal amplification law.

2.4 Group-wise Asymmetric Weight-only Quantization

Local Representation and Residual Error

The purpose of group-wise quantization is to adapt the affine range to local weight statistics. A projection matrix can contain rows and regions with different

minima, maxima, variances, and tail behavior. If all regions share one scale, a few extreme values can enlarge the step size for many otherwise compact weights. Partitioning each output-channel row along input channels reduces the number of weights competing for one range. Asymmetric parameters further permit a local interval that is not centered at zero, while the levels remain uniformly spaced.

This granularity does not eliminate quantization error. Within a group, every weight still uses the same step size and endpoints. A single extreme weight can expand the range of its group. Alternatively, clipping or range selection can preserve finer resolution for the bulk of the group while producing larger endpoint errors for excluded values. The preferable trade-off depends on the objective. Minimizing unweighted element-wise error may favor one range, whereas minimizing calibration-output error may favor another because input features weight the columns unequally.

For a group g with scale s_g and zero-point z_g , adjacent-level adjustment changes a selected integer c_i to $c_i + 1$ or $c_i - 1$, provided the result remains within the representable interval. The corresponding dequantized weight changes by $+s_g$ or $-s_g$. The operation leaves all other integer values and the group’s affine parameters unchanged. This local discreteness makes the effect easy to define, but not automatically beneficial. Moving away from a nearest valid grid point to an adjacent level increases the selected weight’s scalar reconstruction error, including when the nearest point is a clipped endpoint. The only scalar-distance exceptions are an exact tie between neighboring levels or an initial assignment selected by an objective other than nearest scalar reconstruction. Its justification must therefore come from a broader reconstruction or interaction-aware criterion.

The phrase bit-level adjustment is used for this adjacent-level operation throughout the thesis. It does not mean toggling a literal binary bit, changing INT4 to another bit width, or introducing a special high-precision exception. A binary representation is only an encoding of the integer level; the semantic operation is movement to a neighboring integer code. This definition also makes adjustment independent of whether signed integers, offset integers, or another equivalent low-bit encoding is used.

Local and Downstream Objectives

Quantization quality can be measured at several levels. Weight reconstruction compares $\hat{W}$ with W , often using a norm such as $\|\Delta W\|_F^2$. Projection reconstruction evaluates $\|X\Delta W^\top\|_F^2$ on calibration inputs. Attention-aware objectives can compare score matrices, probability matrices, or attention outputs. These objec-

tives are related but not equivalent.

An element-wise objective treats every weight error equally. A projection-output objective weights errors according to the calibration activations and their correlations. A score objective couples Q and K errors because the score matrix is bilinear. An attention-output objective additionally includes softmax sensitivity and value perturbations. Moving to a more downstream objective can better represent the target behavior, but it also becomes more input-dependent and may be more expensive or susceptible to calibration overfitting. A selective refinement method therefore needs an explicitly stated objective and cannot assume that reducing one error measure reduces every later error measure.

The initial group-wise asymmetric quantizer supplies a regular low-bit representation. Selective adjacent-level refinement can then revisit a restricted set of integer assignments under a chosen criterion. Selection is necessary because exhaustively evaluating both neighboring levels for every weight, especially in many large projection matrices, is expensive. Candidate selection may use rounding proximity, local error, distribution summaries, or estimated downstream influence. None of these indicators alone proves that an adjustment will improve model behavior; they narrow the search space for subsequent evaluation.

2.5 Attention Computation and Projection Perturbations

Notation and Projection Orientation

Let $X \in \mathbb{R}^{n \times d_{\text{in}}}$ contain n token representations as rows. Let $W_Q \in \mathbb{R}^{d_Q \times d_{\text{in}}}$, $W_K \in \mathbb{R}^{d_K \times d_{\text{in}}}$, and $W_V \in \mathbb{R}^{d_V \times d_{\text{in}}}$. For a single row vector x , the projections are

$$q = xW_Q^\top, \quad k = xW_K^\top, \quad v = xW_V^\top. \quad (2.3)$$

Applied to all tokens, the corresponding matrices are

$$Q = XW_Q^\top, \quad K = XW_K^\top, \quad V = XW_V^\top. \quad (2.4)$$

This orientation is used consistently: rows of each weight matrix are output channels, columns are input channels, and group-wise quantization partitions columns within each row.

For one attention head, scaled dot-product attention is

$$S = \frac{QK^\top}{\sqrt{d_k}}, \quad P = \text{softmax}_{\text{row}}(S), \quad O = PV, \quad (2.5)$$

where d_k is the head dimension, S is the score matrix, P contains row-wise attention probabilities, and O is the weighted aggregation of values [16]. Multi-head variants apply this process across heads and subsequently combine their outputs.

From Weight Error to Projection Error

Let the quantized projection weights be

$$\hat{W}_Q = W_Q + \Delta W_Q, \quad \hat{W}_K = W_K + \Delta W_K, \quad \hat{W}_V = W_V + \Delta W_V. \quad (2.6)$$

The induced projection perturbations are

$$\Delta Q = X \Delta W_Q^\top, \quad \Delta K = X \Delta W_K^\top, \quad \Delta V = X \Delta W_V^\top, \quad (2.7)$$

so that $\hat{Q} = Q + \Delta Q$, $\hat{K} = K + \Delta K$, and $\hat{V} = V + \Delta V$. The transpose is essential because the rows of each weight matrix are output channels. These equations also make the role of activations explicit: the same ΔW_Q , ΔW_K , or ΔW_V produces different projection errors for different X .

The perturbed unscaled score product expands as

$$\hat{Q} \hat{K}^\top = Q K^\top + \Delta Q K^\top + Q \Delta K^\top + \Delta Q \Delta K^\top. \quad (2.8)$$

After division by $\sqrt{d_k}$, the score perturbation is therefore

$$\Delta S = \frac{\Delta Q K^\top + Q \Delta K^\top + \Delta Q \Delta K^\top}{\sqrt{d_k}}. \quad (2.9)$$

The first two terms are first order in the projection perturbations, while the last term is second order. When perturbations are small, the second-order term may be less influential, but low bit widths, clipping, large activation directions, or aligned errors can make it non-negligible. The expansion shows why independently small Q and K reconstruction errors do not determine score error by themselves: orientation and interaction with the unperturbed projections also matter.

A norm bound illustrates possibilities without predicting realized behavior:

$$\|\Delta S\| \leq \frac{\|\Delta Q\| \|K\| + \|Q\| \|\Delta K\| + \|\Delta Q\| \|\Delta K\|}{\sqrt{d_k}}, \quad (2.10)$$

for compatible submultiplicative norms. This upper bound can be loose because it discards direction and cancellation. It establishes that large projection errors can permit large score error, but it does not prove that the bound is attained or that

QKV weights are universally more sensitive than other model parameters.

Local Sensitivity of Softmax

Softmax converts each score row into a probability distribution. For a score vector s and $p = \text{softmax}(s)$, the Jacobian is

$$J_{\text{softmax}}(s) = \text{diag}(p) - pp^\top. \quad (2.11)$$

For a sufficiently small score perturbation δs , the local first-order probability change is approximately

$$\delta p \approx (\text{diag}(p) - pp^\top) \delta s. \quad (2.12)$$

This expression provides a more precise account than the blanket statement that softmax amplifies errors. Softmax is invariant to adding the same constant to all elements of one score row, so that perturbation direction has no effect. When one probability is already close to one, some perturbations may be strongly attenuated. When several scores are close and receive substantial probability, perturbations that alter their relative differences can redistribute attention meaningfully. The effect is therefore local, directional, and dependent on the current score distribution.

For finite perturbations, a change can cross a region in which the local Jacobian itself changes. A small score error near a narrow decision margin may alter which token receives the largest probability, but even a ranking change need not cause a large output change if the corresponding value vectors are similar. Conversely, a modest probability redistribution can matter if it transfers weight between very different values. Softmax is thus one conditional link in the propagation chain rather than a universal amplifier.

Value Error and Attention-output Error

With $\hat{P} = P + \Delta P$ and $\hat{V} = V + \Delta V$, the attention-output perturbation is

$$\hat{O} - O = \Delta P V + P \Delta V + \Delta P \Delta V. \quad (2.13)$$

The term $\Delta P V$ reflects changes in attention probabilities, $P \Delta V$ reflects direct value-projection error under the original probabilities, and $\Delta P \Delta V$ couples both effects. This decomposition distinguishes the roles of Q/K and V. Q and K affect scores and probabilities; V affects the content being aggregated. A large score perturbation can have limited output effect when relevant values are similar, while a

small value perturbation can matter when it is repeatedly assigned high attention probability.

Later operations, including head concatenation, output projection, residual addition, normalization, and subsequent layers, can attenuate, preserve, or compound the attention-output perturbation. It is therefore defensible to say that QKV quantization creates structured pathways for downstream error, but not that every local QKV error necessarily produces a proportionally larger final error. Claims of comparative sensitivity require direct measurements under specified models, data, quantizers, and evaluation tasks.

2.6 GQA and Dependent Error

GQA changes how query heads are mapped to key and value heads. Suppose query heads $h \in \mathcal{G}_r$ belong to GQA group r and share K_r and V_r . For each associated query head,

$$S_h = \frac{Q_h K_r^\top}{\sqrt{d_k}}, \quad O_h = \text{softmax}_{\text{row}}(S_h) V_r. \quad (2.14)$$

If the shared key is perturbed by ΔK_r , then every $h \in \mathcal{G}_r$ contains the score-error term

$$\frac{Q_h \Delta K_r^\top}{\sqrt{d_k}}. \quad (2.15)$$

These errors are dependent because the same ΔK_r appears in every term. Their realized magnitudes and signs differ because the Q_h differ. Similarly, a shared ΔV_r enters the output of every associated query head, but each head weights that value perturbation with its own attention probabilities.

Dependence matters for analysis because an independence assumption would overlook common error sources. It does not imply coherent accumulation. If query directions differ, some products with ΔK_r may be small or oppose one another after later transformations. If attention distributions differ, the same ΔV_r may be weighted differently. The output projection can mix heads in a manner that cancels or compounds their errors. Consequently, GQA can increase the reach of a shared perturbation and can conditionally amplify its downstream effect, yet the architecture alone does not determine the final magnitude.

Comparisons among MHA, GQA, and MQA should therefore be stated structurally. MHA uses more independent K/V heads; MQA shares K/V most broadly; GQA shares within intermediate-sized architectural groups. Broader sharing creates more query heads that depend on each shared K/V perturbation. Whether this produces worse model quality after quantization depends on the learned weights,

input activations, quantization errors, softmax operating points, and downstream mixing. The thesis focuses on GQA because shared K/V makes interaction-aware analysis especially relevant, not because GQA is assumed to be universally fragile.

The numerical quantization groups used in the thesis remain independent of this architecture. A quantization group lies inside one output-channel row and spans contiguous input channels. A GQA group associates several query heads with a shared key/value head. The two kinds of groups can interact indirectly because quantization error in weights contributing to a shared head is reused architecturally, but neither group definition determines the other.

2.7 Outliers, Saliency, and Candidate Selection

Weight Outliers and Quantization Range

A weight outlier is a weight whose magnitude or value is extreme relative to an explicitly specified reference population, such as its quantization group. The reference population matters: a value can be ordinary at the layer level but extreme within a small local group, or the reverse. Under min–max affine quantization, extreme endpoints influence the scale. Preserving a distant endpoint can enlarge the step size and increase rounding error for many central values. Clipping the endpoint can reduce the step size but incurs a potentially large clipping error for the excluded weight [1], [17].

At INT4, the limited number of levels makes this trade-off visible. Nevertheless, the presence of a weight outlier does not establish that it should be clipped, preserved exactly, or selected for adjustment. Its functional effect depends on the input feature multiplying it, correlations with other errors, and the downstream objective. Group-wise affine quantization localizes the range trade-off, but a single extreme weight can still affect every weight sharing its numerical group.

Activation Outliers and Functional Saliency

An activation outlier is an unusually large or otherwise extreme activation feature under a specified input distribution. It is conceptually distinct from a weight outlier. Weight-only quantization does not directly approximate activations, but activation magnitude influences how weight errors are expressed because projection error equals $X\Delta W^\top$. A modest weight error in a frequently large activation direction can have a substantial projection effect. SmoothQuant [10] and AWQ [3] illustrate, in different settings, why activation statistics are relevant to quantization decisions.

Saliency refers to functional importance under a stated criterion. A weight can

be salient because changing it strongly affects calibration outputs or another objective, even if its magnitude is ordinary. Conversely, a large weight or activation can have limited effect under a particular data distribution or can be compensated elsewhere. Outlier status is distributional; salience is objective-dependent. Treating the terms as synonyms would obscure the difference between selecting extreme values and selecting influential values.

Candidate-selection heuristics can combine these signals, but their roles should remain explicit. Distance to a quantization boundary indicates rounding ambiguity. Magnitude indicates extremeness relative to a reference set. Calibration-weighted error estimates realized projection influence on the calibration sample. Interaction-aware measures estimate effects on score or attention-output objectives. Each signal can narrow the candidate set, but final benefit must be assessed using the selected refinement objective and, ultimately, model evaluation.

Kneedle as a Heuristic

Kneedle was introduced as a general method for locating knee points in suitable curves [6]. A knee is a region where the curve’s rate of improvement or growth changes markedly after normalization and comparison with a reference line. The algorithm’s general literature role is curve-based transition-point identification. It is not inherently an outlier detector, a quantization algorithm, or a proof that a selected threshold is optimal.

In a quantization context, one may construct a monotone summary from sorted weight magnitudes, cumulative energy, reconstruction error, parameter sensitivity, or another quantity and apply Kneedle to propose a transition point. For interaction-aware refinement, a particularly relevant construction is a descending sequence of dimension-level sensitivity values. Sorting separates dimensions by the magnitude of a stated sensitivity measure, while Kneedle proposes where the shape of that ordered sequence changes. The resulting knee can propose a boundary between a small extreme-sensitivity prefix and a subsequent moderate-sensitivity region that remains potentially useful for controlled adjustment. This interpretation depends on the sensitivity definition supplied by the methodology; Kneedle itself neither defines sensitivity nor determines whether any candidate change improves the objective.

The proposed transition also depends on the constructed curve, normalization, curve direction, smoothing, and sensitivity or tolerance parameters. A different summary or normalization can produce a different knee, and some curves may contain no stable knee at all. Applying the algorithm to a sorted dimension-sensitivity

curve can therefore yield a reproducible heuristic boundary or candidate budget, but dimensions before or after that boundary are not thereby proven to be harmful outliers, safe adjustment targets, or functionally salient.

This distinction separates the general literature claim from the later methodology of the thesis. The literature supports Kneedle as a knee-point heuristic. Any use of its output to protect extreme-sensitivity dimensions or to choose moderate-sensitivity candidates for adjacent-level adjustment is a methodological design choice that must be evaluated under the stated quantization objective. The method may help avoid a fixed global threshold when sensitivity profiles differ across heads or groups, but it does not guarantee a better scale, a better candidate set, a globally optimal discrete solution, or improved downstream accuracy.

2.8 Selective Adjacent-level Refinement

After initial group-wise asymmetric quantization, each weight is represented by an integer level and a group-specific affine transformation. A selective refinement stage can evaluate whether moving a candidate integer to an adjacent level improves a chosen objective. If c_i is the stored integer for weight i in group g , the candidate set is restricted to valid neighbors,

$$\mathcal{N}(c_i) = \{c_i - 1, c_i + 1\} \cap [c_{\min}, c_{\max}]. \quad (2.16)$$

Changing c_i to an element of $\mathcal{N}(c_i)$ is the bit-level adjustment considered by this thesis. It moves the dequantized value by exactly one local step s_g . It is not a literal binary-digit flip and does not alter s_g , z_g , group membership, or bit width.

Restricting adjustment to adjacent levels has several conceptual advantages. It bounds the local change, preserves the original quantizer, and permits a clear comparison between the current assignment and its immediate alternatives. It also avoids interpreting a large jump caused by flipping a high-order binary digit as a small refinement. However, adjacency is a constraint, not evidence of improvement. A neighbor can reduce an interaction-aware objective even while increasing element-wise weight error, or it can worsen both. Every accepted adjustment therefore requires an explicit acceptance criterion.

Sequential adjustment also introduces interactions among decisions. The benefit estimated for one candidate can change after another candidate is moved because projection, score, and attention-output errors contain sums and cross terms. Evaluating candidates independently is efficient but may miss coordinated effects. Evaluating all combinations is generally impractical. A selective procedure occupies

the middle ground: use heuristics to restrict candidates, evaluate adjacent alternatives according to a stated objective, and recognize that the resulting solution is a local discrete refinement rather than a proven global optimum.

The choice of objective determines what the refinement preserves. A weight-space objective is inexpensive and input-independent but ignores activation directions. A projection-space objective uses calibration inputs and directly measures $X\Delta W^\top$. A score-space objective couples Q and K but omits the conditional transformation through softmax and values. An attention-output objective includes more of the relevant computation but is more input-dependent. No single objective guarantees improvement on all downstream tasks. The rationale for interaction-aware refinement is therefore that it measures a quantity closer to attention behavior, not that it universally dominates simpler reconstruction.

2.9 Statistical Interpretation and Its Limits

James–Stein Estimation

The James–Stein result concerns estimation of a multivariate normal mean under specific assumptions [7]. In its classical form, one observes

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 I_d), \quad (2.17)$$

with known common variance, squared-error risk, and dimension $d \geq 3$. The James–Stein estimator shrinks the observation vector toward a chosen center, commonly the origin:

$$\hat{\boldsymbol{\theta}}_{\text{JS}} = \left(1 - \frac{(d-2)\sigma^2}{\|\mathbf{x}\|^2}\right) \mathbf{x}. \quad (2.18)$$

Under the classical conditions and the relevant risk definition, shrinkage can dominate the component-wise maximum-likelihood estimator in total expected squared error. The result demonstrates that accepting coordinated bias can reduce aggregate estimation risk in a particular statistical model.

Quantized neural-network weights do not generally satisfy these assumptions. Their errors are discrete, bounded or clipped by a quantizer, grouped under shared affine parameters, multiplied by data-dependent activations, and evaluated through nonlinear model behavior. The unknown-mean Gaussian model, known isotropic variance, and classical squared-error risk do not directly describe this setting. James–Stein theory therefore does not prove that adjacent-level adjustment, Flip, or ReFlip improves a quantized model. It also does not prescribe which weights to adjust or which attention-aware objective to use.

The valid role of James–Stein reasoning in this literature review is modest and analogical. It cautions against assuming that independently optimal component decisions must minimize an aggregate criterion, and it illustrates how controlled bias can improve total risk under suitable assumptions. A representative activation estimate that is shrunk toward a shared center may consequently be described as *James–Stein-inspired*: the inspiration is the stabilization of a multivariate estimate through shrinkage, not the transfer of the classical estimator or its dominance theorem to neural activations. In particular, estimating the shrinkage center or variance from calibration observations changes the statistical setting. Such a heuristic must be justified by its stated construction and evaluated empirically.

James–Stein-inspired shrinkage can therefore motivate a more stable representative activation vector for an activation-weighted objective, but it does not determine the correct shrinkage target, guarantee lower attention-score error, or prove that Flip or ReFlip improves a quantized model. The analogy is useful precisely when its limits are retained: it motivates coordinated estimation and aggregate objectives without replacing quantization-specific analysis.

Connecting Heuristics to Evaluation

Kneedle and James–Stein reasoning occupy different roles. Kneedle is an algorithmic heuristic for proposing a transition point on a constructed curve. James–Stein is a statistical theorem under a specific probabilistic model that offers a limited analogy about aggregate risk. Neither establishes that a selected quantization candidate is salient, that an adjacent-level move is beneficial, or that a downstream model metric will improve.

A defensible refinement framework therefore separates proposal, acceptance, and validation. A heuristic may propose candidate weights or a candidate budget. A specified reconstruction or interaction-aware objective determines whether an adjacent-level alternative is accepted. Model-level evaluation determines whether the resulting quantized model preserves behavior on relevant data. This separation prevents a selection heuristic from being interpreted as proof and prevents an analogy from being interpreted as an optimization guarantee.

2.10 Synthesis and Research Gap

The literature establishes several relevant facts while leaving a focused gap. First, post-training quantization can compress large language models without full retraining, and AdaRound, GPTQ, and AWQ demonstrate distinct ways to use reconstruction objectives, approximate curvature, and activation statistics rather than

relying only on independent scalar-nearest assignments [3], [12], [13]. Second, affine asymmetry and group-wise granularity address different aspects of representation: asymmetry shifts a uniformly spaced local range, while grouping localizes the parameters shared by weights. Third, attention couples projection errors through a bilinear score product, a locally sensitive softmax transformation, and value aggregation. Fourth, GQA causes several query heads to depend on shared K/V perturbations, creating dependent errors whose eventual amplification, attenuation, or cancellation is conditional.

The evidence does not support several stronger claims. KV-cache studies do not directly prove projection-weight sensitivity. Weight magnitude does not establish salience. Softmax does not universally amplify score perturbations. GQA sharing does not guarantee coherent accumulation. Kneedle does not prove that a threshold separates harmful outliers, and James–Stein theory does not prove the value of a discrete quantization refinement. Preserving these boundaries narrows the thesis claim but makes it testable.

Within these boundaries, the research gap is not adaptive rounding in general. It is the selective refinement of group-wise asymmetric weight-only low-bit QKV projection weights through two narrowly defined correction mechanisms after initial quantization. The first evaluates adjacent integer-code changes against an aggregate activation-weighted projection residual. The second restricts correction to query weights and evaluates adjacent changes against a scalar query–key interaction surrogate within a GQA setting. Sorted dimension sensitivity and Kneedle can restrict the candidate region, but exact objective changes determine acceptance. These objectives and restrictions distinguish the proposed refinement from RTN, general layer-reconstruction methods, and channel-scaling methods without implying universal superiority.

The initial quantizer provides a regular, efficient representation, while bit-level adjustment provides a restricted discrete action after quantization. The resulting procedures remain local searches: they do not alter the bit width or affine parameters, do not guarantee a global optimum, and do not ensure improvement on every downstream metric. Their motivation follows from the mismatch between scalar-nearest assignments and aggregate or interaction-aware objectives; their usefulness must be established empirically.

Conclusion

This chapter has defined the thesis setting as post-training, group-wise asymmetric INT4 weight-only quantization of QKV projection weights. It distinguished

input channels, output channels, numerical quantization groups, and architectural GQA groups; clarified that asymmetric affine quantization remains uniformly spaced; and defined bit-level adjustment as movement to an adjacent integer code and quantization level rather than a literal binary-digit flip.

The perturbation analysis showed that projection-weight errors induce $\Delta Q = X\Delta W_Q^\top$, $\Delta K = X\Delta W_K^\top$, and $\Delta V = X\Delta W_V^\top$. Query and key errors interact in the score product, after which softmax transforms score perturbations according to its local, directional sensitivity. GQA reuses shared K/V perturbations across associated query heads and therefore creates dependent errors, but their downstream accumulation or amplification remains conditional on queries, score margins, values, output mixing, and later computations.

Finally, the chapter established careful evidence boundaries. KVQuant and QAO provide adjacent evidence about the KV cache rather than direct proof about projection weights. AdaRound, GPTQ, and AWQ are structured post-training methods, but their objectives differ from the two explicit correction objectives considered here. Kneedle can guide selection on a sorted sensitivity curve, but it does not certify outliers, optimal thresholds, or beneficial adjustments. James–Stein estimation motivates shrinkage only by analogy. A James–Stein-inspired activation estimate does not inherit the classical dominance guarantee. These conclusions motivate the later methodology while leaving its effectiveness to direct empirical evaluation.

CHAPTER 3. METHODOLOGY

3.1 Overview

This chapter presents a post-training, weight-only quantization methodology for the query, key, and value projections of transformer attention. The methodology is designed for asymmetric group-wise INT4 quantization: projection weights are stored on a four-bit integer grid with group-specific scales and zero points, whereas activations remain in their original precision [2], [11]. The central premise is that nearest rounding is a strong starting point but not necessarily the best final choice when the quantized weights are evaluated through the activations and interactions that determine attention.

The proposed refinement has two complementary stages. The first stage, called **Flip**, treats a projection matrix independently and reduces its activation-weighted output residual. Flip is deliberately distinguished from ordinary round-to-nearest (RTN) quantization. RTN already minimizes the isolated scalar reconstruction error of each weight on a fixed quantization grid. Consequently, changing a rounded integer to a neighboring level generally cannot improve that isolated scalar objective. Flip instead accepts a neighboring level only when the resulting change reduces the aggregate projected residual. Such a change may increase the error of an individual weight while improving the output of the complete projection through cancellation among activation-weighted errors.

The second stage, called **ReFlip**, addresses the coupling between query and key projections within grouped-query attention (GQA). For each query head, ReFlip holds its group’s shared quantized key head fixed and updates only that query head’s quantized query matrix. It evaluates how a legal one-step integer change affects an unscaled representative-vector query–key surrogate. The resulting objective is exact for this scalar surrogate: no Taylor approximation or full-matrix gradient surrogate is used. ReFlip can therefore compensate for a surrogate residual jointly caused by query and key quantization, even though only query weights are modified. This compensation is useful, but its scope must be stated precisely: preservation of one representative scalar surrogate does not imply preservation of the full pairwise attention-score matrix over all token pairs.

Both refinement stages use a representative activation row vector obtained from calibration data and stabilized by James–Stein shrinkage. This produces a compact, data-dependent signal for evaluating discrete changes without retraining. Kneedle-based selection is used in ReFlip to identify an extreme high-sensitivity region that

should be protected and a moderate-sensitivity region immediately after the knee in which correction is permitted. Kneedle is an adaptive heuristic rather than an optimality guarantee.

The overall procedure remains fully post-training. It requires neither gradient-based optimization nor modification of the model architecture. The refinement is performed before deployment, and the resulting matrices retain the same group-wise asymmetric INT4 representation used by the initial quantizer. Thus, the method aims to improve the functional fidelity of quantized attention projections while preserving the principal memory motivation of low-bit weight storage and without making claims about measured runtime.

3.2 Notation and Calibration Representative

Projection Convention

Let a representative activation be a row vector

$$x \in \mathbb{R}^{d_{\text{in}}}. \quad (3.1)$$

For a generic projection, let the full-precision weight matrix have shape

$$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, \quad (3.2)$$

so that the projected row vector is

$$y = xW^\top \in \mathbb{R}^{d_{\text{out}}}. \quad (3.3)$$

This row-vector convention is used throughout the chapter. In particular, matrix row a identifies an output dimension and matrix column b identifies an input dimension.

The quantized and dequantized approximation is denoted by

$$\hat{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, \quad (3.4)$$

with projected vector

$$\hat{y} = x\hat{W}^\top. \quad (3.5)$$

Although the stored representation is integer-valued, the hat notation refers to the dequantized value represented by those integers, scales, and zero points.

For GQA [5], let $\mathcal{H}_Q$ be the set of query-head indices and $\mathcal{H}_{KV}$ the set of

key/value-head indices. The query heads are partitioned into GQA groups

$$\{\mathcal{G}_r : r \in \mathcal{H}_{KV}\}, \quad \bigcup_{r \in \mathcal{H}_{KV}} \mathcal{G}_r = \mathcal{H}_Q, \quad \mathcal{G}_r \cap \mathcal{G}_{r'} = \emptyset \quad (r \neq r'), \quad (3.6)$$

where every query head $h \in \mathcal{G}_r$ shares key/value head r . With common head dimension d_h , the head-specific projection matrices are

$$W_Q^{(h)}, W_K^{(r)}, W_V^{(r)} \in \mathbb{R}^{d_h \times d_{\text{in}}}, \quad h \in \mathcal{G}_r. \quad (3.7)$$

Their representative projected vectors are

$$q^{(h)} = x(W_Q^{(h)})^\top, \quad k^{(r)} = x(W_K^{(r)})^\top, \quad v^{(r)} = x(W_V^{(r)})^\top, \quad (3.8)$$

and the corresponding dequantized vectors are written $\hat{q}^{(h)}$, $\hat{k}^{(r)}$, and $\hat{v}^{(r)}$. This notation distinguishes a GQA group, which describes head sharing, from a quantization group, which describes columns sharing quantization parameters.

James–Stein–Shrunk Activation

Calibration activations can be noisy when only a limited post-training sample is available. A single unregularized statistic may overemphasize accidental extremes, whereas a simple global average may suppress meaningful differences between input dimensions. To obtain a regularized representative row vector, this work uses a nonnegative James–Stein-inspired shrinkage heuristic, while retaining the distinction between this calibration construction and the assumptions of the classical estimator [7].

Let $X^{(1)}, \dots, X^{(N_{\text{cal}})} \in \mathbb{R}^{d_{\text{in}}}$ denote the calibration activation rows. The empirical channel-mean vector and its grand mean are

$$\tilde{x}_b = \frac{1}{N_{\text{cal}}} \sum_{n=1}^{N_{\text{cal}}} X_b^{(n)}, \quad \bar{x} = \frac{1}{d_{\text{in}}} \sum_{b=1}^{d_{\text{in}}} \tilde{x}_b. \quad (3.9)$$

The shrinkage target is the constant vector $m = \bar{x}\mathbf{1}$, and the calibration-noise proxy is the squared mean absolute deviation of the channel means,

$$\hat{\sigma}^2 = \left(\frac{1}{d_{\text{in}}} \sum_{b=1}^{d_{\text{in}}} |\tilde{x}_b - \bar{x}| \right)^2. \quad (3.10)$$

The representative activation is

$$x = m + \lambda(\tilde{x} - m), \quad (3.11)$$

where the positive-part shrinkage factor has the form

$$\lambda = \left[1 - \frac{(d_{\text{in}} - 2)\hat{\sigma}^2}{\|\tilde{x} - m\|_2^2} \right]_+, \quad [u]_+ = \max(u, 0). \quad (3.12)$$

If $\|\tilde{x} - m\|_2^2 = 0$, the representative is defined directly as $x = \tilde{x} = m$ and no shrinkage calculation is required. Otherwise, if the observed dimension-wise variation is weak relative to the noise proxy, λ moves the statistic toward m ; if the observed variation is strong, more of the empirical structure is retained. Because both m and $\hat{\sigma}^2$ are estimated from the same calibration observations and the noise proxy is heuristic, this construction is inspired by the classical estimator rather than an invocation of its classical risk-dominance theorem. The resulting x is not claimed to represent every token or every sequence. It is a compact calibration summary used to define the exact objectives evaluated by Flip and ReFlip.

This distinction is important. The refinement stages target behavior under the chosen representative activation. Their objectives can improve a targeted calibration criterion, but they do not provide a universal guarantee over the complete activation distribution.

3.3 Problem Formulation

The goal is to construct group-wise asymmetric INT4 approximations of the attention projection matrices while limiting functionally relevant quantization error. Conventional RTN quantization primarily considers each scalar weight on its own. Attention, however, is determined by projected activations and by interactions between the resulting query and key vectors.

For a generic projection matrix W , define the dequantization error

$$E = \hat{W} - W. \quad (3.13)$$

Under the representative activation, the corresponding projection residual is

$$r = xE^\top = x(\hat{W} - W)^\top \in \mathbb{R}^{d_{\text{out}}}. \quad (3.14)$$

The residual in output dimension j is therefore

$$r_j = \sum_{i=1}^{d_{\text{in}}} x_i E_{ji}. \quad (3.15)$$

This expression makes the relevant error aggregation explicit. Individual signed weight errors are multiplied by activation coordinates and then summed. Two nonzero scalar weight errors can partially cancel in the projected output, while a small scalar error can matter substantially when multiplied by a large activation coordinate.

The local functional objective used by Flip is

$$J(W, \hat{W}; x) = \|r\|_2^2 = \|x(\hat{W} - W)^\top\|_2^2. \quad (3.16)$$

This objective is applicable independently to W_Q , W_K , and W_V . It measures projection fidelity for the representative activation and does not claim to optimize the full attention computation.

The interaction-aware objective used by ReFlip is narrower and more explicit. For query head $h \in \mathcal{G}_r$, define the original unscaled representative-vector query-key surrogate

$$s_{h,r}^* = \left[x(W_Q^{(h)})^\top \right] \cdot \left[x(W_K^{(r)})^\top \right] = q^{(h)} \cdot k^{(r)}, \quad (3.17)$$

and the current quantized surrogate

$$\hat{s}_{h,r} = \left[x(\hat{W}_Q^{(h)})^\top \right] \cdot \left[x(\hat{W}_K^{(r)})^\top \right] = \hat{q}^{(h)} \cdot \hat{k}^{(r)}. \quad (3.18)$$

ReFlip uses the per-query-head loss

$$\mathcal{L}_{h,r} = \frac{1}{2} (\hat{s}_{h,r} - s_{h,r}^*)^2. \quad (3.19)$$

The scalar residual

$$e_{h,r} = \hat{s}_{h,r} - s_{h,r}^* \quad (3.20)$$

contains the combined effect of quantizing query head h and its shared key head r . ReFlip holds $\hat{W}_K^{(r)}$ fixed and changes only $\hat{W}_Q^{(h)}$, using the latter as the available degree of freedom for reducing this joint surrogate residual. The word “unscaled” is deliberate: this compact surrogate omits the usual $1/\sqrt{d_h}$ factor because multiplication by a positive constant would not change which exact one-step candidate decreases the squared residual. It is not presented as a general or complete attention

logit.

These two objectives establish a progression from projection-level fidelity to query–key interaction fidelity. Flip targets an activation-weighted residual for one matrix at a time. ReFlip then compensates the remaining per-query-head scalar Q–K surrogate residual through legal integer changes to $\hat{W}_Q^{(h)}$. Neither objective is presented as an exact surrogate for all downstream model behavior.

3.4 Group-Wise Asymmetric INT4 Quantization

Integer Representation

For a generic projection matrix

$$W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, \quad (3.21)$$

the input dimension is partitioned into contiguous quantization groups of at most G columns. Let $\gamma(b)$ map input column b to its quantization-group index. Each output row and quantization group has a scale

$$\alpha_{a,\gamma(b)} > 0 \quad (3.22)$$

and a zero point

$$\zeta_{a,\gamma(b)} \in \mathcal{Z}_4. \quad (3.23)$$

The stored INT4 value at row a and column b is

$$z_{ab} \in \mathcal{Z}_4 = \{0, 1, \dots, 15\}. \quad (3.24)$$

Its dequantized value is

$$\hat{W}_{ab} = \alpha_{a,\gamma(b)} (z_{ab} - \zeta_{a,\gamma(b)}). \quad (3.25)$$

For row a and quantization group t , let $\mathcal{I}_t = \{b : \gamma(b) = t\}$ be its input-column set. The real range is enlarged to include zero:

$$w_{a,t}^- = \min \left(0, \min_{b \in \mathcal{I}_t} W_{ab} \right), \quad w_{a,t}^+ = \max \left(0, \max_{b \in \mathcal{I}_t} W_{ab} \right). \quad (3.26)$$

When the augmented range is nondegenerate, the asymmetric parameters are

$$\alpha_{a,t} = \frac{w_{a,t}^+ - w_{a,t}^-}{15}, \quad \zeta_{a,t} = \text{clip}_{[0,15]} \left(\text{round} \left(-\frac{w_{a,t}^-}{\alpha_{a,t}} \right) \right). \quad (3.27)$$

Including zero handles one-sided groups explicitly. If all weights in a group are nonnegative, then $w_{a,t}^- = 0$ and $\zeta_{a,t} = 0$; if all are nonpositive, then $w_{a,t}^+ = 0$ and the zero point is placed at the upper end of the integer range, up to the stated rounding and clipping rule. For a mixed-sign group, the zero point lies inside the legal range.

The initial RTN integer is

$$z_{ab}^{\text{RTN}} = \text{clip}_{[0,15]} \left(\text{round} \left(\frac{W_{ab}}{\alpha_{a,\gamma(b)}} + \zeta_{a,\gamma(b)} \right) \right). \quad (3.28)$$

Under the zero-inclusive range, a degenerate group satisfies $w_{a,t}^- = w_{a,t}^+ = 0$ and therefore contains only zeros. Such a group is represented exactly by the explicit convention

$$\alpha_{a,t} = 1, \quad \zeta_{a,t} = 0, \quad z_{ab} = 0 \quad \text{for all } b \in \mathcal{I}_t, \quad (3.29)$$

and is frozen during refinement because every useful target value in the group is already represented exactly. This convention avoids division by zero without assigning an arbitrary effective step to an active nonzero group.

What RTN Optimizes

For fixed $\alpha_{a,\gamma(b)}$ and $\zeta_{a,\gamma(b)}$, RTN chooses the legal quantization level closest to W_{ab} . Including legal endpoint levels, it solves the isolated scalar problem, with non-uniqueness only at exact ties:

$$\min_{z \in \mathcal{Z}_4} \left| \alpha_{a,\gamma(b)} (z - \zeta_{a,\gamma(b)}) - W_{ab} \right|. \quad (3.30)$$

Equivalently, it minimizes the squared scalar reconstruction error on the fixed grid. Therefore, a neighboring integer change should not be described as improving the isolated scalar weight error that RTN already minimizes.

The limitation of RTN is instead an objective mismatch. The projected residual in output row a depends on the signed sum

$$r_a = \sum_b x_b (\hat{W}_{ab} - W_{ab}), \quad (3.31)$$

not on each absolute scalar error separately. A neighboring integer can increase

$$|\hat{W}_{ab} - W_{ab}| \quad (3.32)$$

yet reduce $|r_a|$ because its signed activation-weighted contribution better cancels the contributions of other weights. Flip exploits precisely this possibility.

Memory and Inference Motivation

INT4 storage provides the theoretical basis for reducing the weight memory of QKV projections. Ignoring metadata, four-bit weights require one quarter of the storage of FP16 weights and one eighth of the storage of FP32 weights. Group-wise scales and zero points introduce overhead, so the exact compression ratio depends on group size and metadata precision. Smaller groups can improve local representational fidelity but increase this overhead.

The refinement stages do not change the deployed tensor shapes or the mathematical dequantization rule. They modify legal stored integers while retaining the same scales and zero points. Consequently, the refined model remains compatible with the same class of group-wise weight-only inference procedure as its RTN initialization. Because the present methodology does not report a measured runtime study, it makes no empirical latency or throughput claim. Its inference motivation is structural: low-bit weight storage can reduce memory traffic and capacity requirements, while the offline refinement seeks to recover fidelity without introducing a new runtime objective or a new projection operator.

3.5 Flip: Activation-Weighted Projection Refinement

Flip Objective

For one projection matrix W and its current quantized approximation $\hat{W}$, Flip targets

$$J = \|r\|_2^2, \quad r = xE^\top, \quad E = \hat{W} - W. \quad (3.33)$$

The objective separates across output rows:

$$J = \sum_{j=1}^{d_{\text{out}}} r_j^2. \quad (3.34)$$

Nevertheless, it does not separate across the weights within a row because all input coordinates contribute to the same scalar residual r_j . This within-row coupling is the source of useful error cancellation.

Suppose a legal candidate changes only the dequantized weight at row j and column i by

$$\delta = \hat{W}'_{ji} - \hat{W}_{ji}. \quad (3.35)$$

For a one-step integer move,

$$\delta = \alpha_{j,\gamma(i)} \Delta z, \quad \Delta z \in \{-1, +1\}, \quad (3.36)$$

provided that $z_{ji} + \Delta z \in \mathcal{Z}_4$. The candidate changes only residual component r_j :

$$r'_j = r_j + x_i \delta. \quad (3.37)$$

All other residual components remain unchanged. The exact objective change is consequently

$$\Delta J = J' - J \quad (3.38)$$

$$= (r_j + x_i \delta)^2 - r_j^2 \quad (3.39)$$

$$= 2r_j x_i \delta + (x_i \delta)^2. \quad (3.40)$$

No approximation is required. A legal candidate is accepted only if

$$\Delta J < 0. \quad (3.41)$$

After acceptance, the scalar residual for that output row is updated as

$$r_j \leftarrow r_j + x_i \delta. \quad (3.42)$$

This incremental update preserves exact agreement with the current quantized matrix while avoiding repeated reconstruction of the full residual.

Interpretation as Error Cancellation

The sign of the first term in ΔJ determines whether a candidate initially moves against the current residual:

$$2r_j x_i \delta < 0. \quad (3.43)$$

The second term,

$$(x_i \delta)^2, \quad (3.44)$$

penalizes an excessively large move and prevents a candidate from being accepted merely because it has the correct direction. The exact acceptance condition can also be written as

$$|r_j + x_i \delta| < |r_j|. \quad (3.45)$$

Thus, Flip selects a neighboring quantization level only when its activation-weighted contribution reduces the magnitude of the aggregate row residual.

This interpretation resolves an apparent contradiction between RTN and Flip. RTN is already optimal for the isolated scalar reconstruction error on the fixed grid. Flip does not overturn that fact. Instead, it changes the objective from isolated

scalar error to activation-weighted projection error. Consider a row containing two signed weight errors whose activation-weighted contributions reinforce one another. Moving one weight away from its nearest level can reverse or weaken one contribution, producing greater cancellation in their sum. The modified weight can be less accurate in isolation while the projected output becomes more accurate.

Accordingly, Flip must not be characterized as a second nearest-rounding search. It is a discrete coordinate refinement under the exact aggregate objective J . Its benefit depends on the representative activation and on the signed configuration of errors within each output row.

Candidate Evaluation and Acceptance

Flip begins from the RTN-quantized matrix and its exact residual r . For each eligible integer z_{ji} , it evaluates the legal directions in

$$\mathcal{D}_{ji} = \{\Delta z \in \{-1, +1\} : z_{ji} + \Delta z \in \mathcal{Z}_4\}. \quad (3.46)$$

Here an eligible coordinate is one whose quantization group is nondegenerate and not frozen, and whose legal direction set $\mathcal{D}_{ji}$ is nonempty. Frozen degenerate groups and coordinates with no legal one-level move are skipped rather than assigned an artificial step size. For each direction, it obtains $\delta = \alpha_{j,\gamma(i)}\Delta z$ and evaluates the exact ΔJ . If neither direction decreases J , the integer remains unchanged. When both symmetric directions are legal and $x_i\alpha_{j,\gamma(i)} \neq 0$, at most one can strictly reduce J : adding their two exact objective changes yields the positive quantity $2(x_i\alpha_{j,\gamma(i)})^2$. Thus, the method accepts the unique strictly improving direction when one exists. Once a change is accepted, both z_{ji} and r_j are updated before subsequent candidates are evaluated.

Updating the residual after every accepted change is essential. Candidate gains are not independent: a previous change alters the cancellation needed in row j , and therefore alters the exact value of every later ΔJ in that row. Evaluating all candidates against a stale initial residual would not implement the stated greedy objective.

Candidate order is a heuristic choice. One option is to evaluate coordinates in descending order of the magnitude of their possible activation-weighted step,

$$|x_i|\alpha_{j,\gamma(i)}. \quad (3.47)$$

Another is to scan deterministically by row and column. The acceptance equation

remains exact under either ordering, but the final discrete configuration can differ because accepted changes modify later residuals. A pass budget and an acceptance budget are therefore explicit hyperparameters rather than hidden assumptions.

Scope and Limitations of Flip

Flip can be applied independently to the query, key, and value projection matrices because its objective requires only a matrix, its quantized approximation, and the representative activation. For W_Q and W_K , reducing the projection residual can indirectly improve the vectors used in query–key scoring. For W_V , it can improve the representative value projection. However, Flip does not directly optimize the interaction between query and key, the softmax operation, or the final attention output.

The objective is also calibration-dependent. A change that reduces J under the representative James–Stein-shrunk vector may not reduce projection error for every activation row. The shrinkage estimate and conservative integer neighborhood are intended to limit sensitivity to calibration noise, but they do not constitute a distribution-wide guarantee. Finally, the greedy coordinate procedure can stop at a discrete local configuration even when a coordinated multi-weight change would yield a lower objective.

3.6 ReFlip: Scalar Query–Key Surrogate Refinement

Per-Query-Head Fixed-Key Objective

After initial quantization and the Flip stage in the proposed refinement sequence, ReFlip treats each query head $h \in \mathcal{G}_r$ separately while using the fixed shared key head r . The full-precision and current quantized unscaled representative-vector Q–K surrogates are

$$s_{h,r}^* = q^{(h)} \cdot k^{(r)} \quad (3.48)$$

and

$$\hat{s}_{h,r} = \hat{q}^{(h)} \cdot \hat{k}^{(r)}. \quad (3.49)$$

Define the per-head scalar residual and loss

$$e_{h,r} = \hat{s}_{h,r} - s_{h,r}^*, \quad \mathcal{L}_{h,r} = \frac{1}{2} e_{h,r}^2. \quad (3.50)$$

During the correction of query head h , $\hat{W}_K^{(r)}$ is fixed. ReFlip updates $\hat{W}_Q^{(h)}$ only; it does not update any key or value matrix. Although the available changes belong only to one query head, $e_{h,r}$ includes quantization effects from both that query

head and its shared key head. ReFlip may therefore use a query-weight change to compensate part of the joint Q–K surrogate residual.

This scalar is explicitly an unscaled representative-vector surrogate. It is narrower than a transformer’s complete attention computation: it uses one calibration representative on both projection paths, omits cross-token query–key pairs, and does not include softmax or value aggregation. Consequently, reducing $\mathcal{L}_{h,r}$ does not assert preservation of the full attention-score matrix or of a general attention logit.

Exact Effect of an Integer Change

Consider integer $z_{ab}^{(h)}$ representing query-head weight (a, b) . ReFlip directly changes

$$z_{ab}^{(h)} \leftarrow z_{ab}^{(h)} + \Delta z_{ab}^{(h)}, \quad \Delta z_{ab}^{(h)} \in \{-1, +1\}, \quad (3.51)$$

subject to

$$z_{ab}^{(h)} + \Delta z_{ab}^{(h)} \in \mathbb{Z}_4. \quad (3.52)$$

Because the query-head scale and zero point remain fixed, the dequantized weight change is

$$\Delta \hat{W}_{Q,ab}^{(h)} = \alpha_{a,\gamma(b)}^{(h)} \Delta z_{ab}^{(h)}. \quad (3.53)$$

Only query coordinate $\hat{q}_a^{(h)}$ changes:

$$\Delta \hat{q}_a^{(h)} = x_b \alpha_{a,\gamma(b)}^{(h)} \Delta z_{ab}^{(h)}. \quad (3.54)$$

Since $\hat{k}^{(r)}$ is fixed, the exact surrogate change is

$$\Delta s_{ab}^{(h,r)} = \hat{k}_a^{(r)} x_b \alpha_{a,\gamma(b)}^{(h)} \Delta z_{ab}^{(h)}. \quad (3.55)$$

This expression is exact for the stated scalar surrogate, rather than a first-order approximation. Its exact loss change is

$$\Delta \mathcal{L}_{ab}^{(h,r)} = \frac{1}{2} \left(e_{h,r} + \Delta s_{ab}^{(h,r)} \right)^2 - \frac{1}{2} e_{h,r}^2 \quad (3.56)$$

$$= e_{h,r} \Delta s_{ab}^{(h,r)} + \frac{1}{2} \left(\Delta s_{ab}^{(h,r)} \right)^2. \quad (3.57)$$

A candidate is accepted only if

$$\Delta \mathcal{L}_{ab}^{(h,r)} < 0. \quad (3.58)$$

After acceptance, the integer parameter, dequantized query weight, query coordinate, scalar surrogate, and residual are updated immediately:

$$z_{ab}^{(h)} \leftarrow z_{ab}^{(h)} + \Delta z_{ab}^{(h)}, \quad \hat{W}_{Q,ab}^{(h)} \leftarrow \hat{W}_{Q,ab}^{(h)} + \Delta \hat{W}_{Q,ab}^{(h)}, \quad (3.59)$$

$$\hat{q}_a^{(h)} \leftarrow \hat{q}_a^{(h)} + \Delta \hat{q}_a^{(h)}, \quad \hat{s}_{h,r} \leftarrow \hat{s}_{h,r} + \Delta s_{ab}^{(h,r)}, \quad (3.60)$$

$$e_{h,r} \leftarrow e_{h,r} + \Delta s_{ab}^{(h,r)}. \quad (3.61)$$

The immediate residual update is necessary because every subsequent exact loss change is evaluated against the current quantized state.

Integer-Level and Dimension Sensitivity

The magnitude of surrogate change per unit change in the integer parameter directly modified by ReFlip defines the integer-level sensitivity:

$$\mathcal{S}_{ab}^{(h,r)} = \left| \frac{\Delta s_{ab}^{(h,r)}}{\Delta z_{ab}^{(h)}} \right| = \left| \hat{k}_a^{(r)} x_b \alpha_{a,\gamma(b)}^{(h)} \right|. \quad (3.62)$$

This quantity combines the participation of key coordinate a in the fixed shared head, the representative activation coordinate b , and the actual dequantized distance associated with a one-level query-integer move. Including the group scale is essential: $|\hat{k}_a^{(r)} x_b|$ alone is not the sensitivity with respect to the integer parameter.

Let $\mathcal{B}_a^{(h)}$ contain the input columns whose query weights belong to nondegenerate quantization groups and admit at least one legal one-level move. Frozen or otherwise ineligible coordinates are assigned zero sensitivity and excluded from candidate-region construction. Integer-level sensitivities are then aggregated over eligible input columns with an ℓ_2 norm:

$$\mathcal{S}_a^{(h,r)} = \left(\sum_{b \in \mathcal{B}_a^{(h)}} \left(\mathcal{S}_{ab}^{(h,r)} \right)^2 \right)^{1/2} \quad (3.63)$$

$$= \left| \hat{k}_a^{(r)} \right| \left(\sum_{b \in \mathcal{B}_a^{(h)}} \left(\alpha_{a,\gamma(b)}^{(h)} \right)^2 x_b^2 \right)^{1/2}. \quad (3.64)$$

This dimension-level quantity is an ℓ_2 -aggregated sensitivity heuristic used to choose where to search. It is not the exact benefit of a legal move and is not an optimal correction-capacity measure. Every admitted integer candidate must still satisfy the exact loss-decrease condition.

Kneedle-Based Protection and Candidate Region

For each pair (h, r) , let $\pi_{h,r}(t)$ be the original query-output index at sorted position t , with sensitivities ordered decreasingly:

$$\mathcal{S}_{\pi_{h,r}(1)}^{(h,r)} \geq \mathcal{S}_{\pi_{h,r}(2)}^{(h,r)} \geq \dots \geq \mathcal{S}_{\pi_{h,r}(d_h)}^{(h,r)}. \quad (3.65)$$

Before applying Kneedle [6], both axes are normalized to the unit interval. For $d_h > 1$, define normalized rank

$$u_t = \frac{t - 1}{d_h - 1}, \quad t = 1, \dots, d_h, \quad (3.66)$$

and, when the sensitivity range is nonzero, normalized decreasing sensitivity

$$y_t = \frac{\mathcal{S}_{\pi_{h,r}(t)}^{(h,r)} - \mathcal{S}_{\pi_{h,r}(d_h)}^{(h,r)}}{\mathcal{S}_{\pi_{h,r}(1)}^{(h,r)} - \mathcal{S}_{\pi_{h,r}(d_h)}^{(h,r)}}. \quad (3.67)$$

Kneedle is configured for a decreasing, convex curve (u_t, y_t) with sensitivity parameter κ . Let $\rho_{h,r}$ denote the detected knee position. The knee itself is included in the protected extreme region:

$$\mathcal{P}_{h,r} = \{\pi_{h,r}(1), \dots, \pi_{h,r}(\rho_{h,r})\}. \quad (3.68)$$

With moderate-region width H_R , the allowed query-output rows are

$$\mathcal{M}_{h,r} = \{\pi_{h,r}(\rho_{h,r} + 1), \dots, \pi_{h,r}(\min(\rho_{h,r} + H_R, d_h))\}. \quad (3.69)$$

This definition makes the normalization, curve direction, and treatment of the knee explicit. If $d_h \leq 1$, all dimension sensitivities are equal, the normalized sensitivity denominator is zero, or Kneedle reports no knee under the selected κ , the conservative fallback is to skip ReFlip for query head h . The fallback avoids inventing an arbitrary moderate region when the sensitivity curve supplies no defensible transition.

Kneedle determines only a candidate region. It does not certify that the protected set, moderate set, or final integer configuration is optimal. Within $\mathcal{M}_{h,r}$, integer-level sensitivities may order candidates, but the exact $\Delta\mathcal{L}_{ab}^{(h,r)}$ remains the sole acceptance criterion.

Greedy ReFlip Procedure

ReFlip proceeds query head by query head. For $h \in \mathcal{G}_r$, it begins from current $\hat{W}_Q^{(h)}$, fixed $\hat{W}_K^{(r)}$, and exact residual $e_{h,r}$. It computes integer-level and dimension-level sensitivities, applies the normalized decreasing-curve Kneedle rule, protects $\mathcal{P}_{h,r}$, and searches only $\mathcal{M}_{h,r}$ when a valid knee exists.

For each legal candidate $(a, b, \Delta z_{ab}^{(h)})$, the method evaluates

$$\Delta s_{ab}^{(h,r)} = \hat{k}_a^{(r)} x_b \alpha_{a,\gamma(b)}^{(h)} \Delta z_{ab}^{(h)} \quad (3.70)$$

and

$$\Delta \mathcal{L}_{ab}^{(h,r)} = e_{h,r} \Delta s_{ab}^{(h,r)} + \frac{1}{2} \left(\Delta s_{ab}^{(h,r)} \right)^2. \quad (3.71)$$

Candidates with nonnegative exact loss change are rejected. When both symmetric directions are legal and $\mathcal{S}_{ab}^{(h,r)} > 0$, at most one can strictly reduce the quadratic loss because the sum of the two exact loss changes is $\left(\mathcal{S}_{ab}^{(h,r)} \right)^2 > 0$. The unique improving direction, when it exists, is accepted and all affected current-state quantities are updated before another candidate is evaluated. The procedure stops at its accepted-change budget, its pass budget, or a pass with no accepted candidate.

The exact acceptance condition has the equivalent form

$$\left| e_{h,r} + \Delta s_{ab}^{(h,r)} \right| < |e_{h,r}|. \quad (3.72)$$

Thus, ReFlip accepts a change only when it moves the current per-head surrogate closer to its full-precision scalar target. The quadratic term rejects a step that overshoots the target by more than it corrects.

Scope and Limitations of ReFlip

ReFlip directly targets a query–key interaction that Flip does not model. A residual caused partly by $\hat{W}_K^{(r)} - W_K^{(r)}$ can be reduced by a compensating change in $\hat{W}_Q^{(h)}$. This is a functional correction: the selected query weight need not become closer to its full-precision value. In particular, an accepted ReFlip change may increase the query-head projection residual

$$J_Q^{(h)} = \left\| x \left(\hat{W}_Q^{(h)} - W_Q^{(h)} \right)^\top \right\|_2^2 \quad (3.73)$$

while strictly reducing $\mathcal{L}_{h,r}$. The two objectives measure different quantities, so ReFlip makes no monotonicity claim for $J_Q^{(h)}$.

The candidate surrogate change and loss change are exact only for the defined unscaled representative-vector scalar. ReFlip does not directly preserve scores between distinct token rows, scaled logits, the softmax distribution, or the value-weighted attention output. A lower per-head scalar loss is evidence only for that targeted criterion.

ReFlip is also greedy and discrete. Candidate ordering and the Kneedle-selected region can affect the endpoint, and a sequence of individually rejected changes may be beneficial if applied jointly. Kneedle does not remove this limitation and does not provide an optimality certificate. ReFlip therefore guarantees strict local improvement only for every accepted change under the stated scalar objective; it does not guarantee a globally optimal query matrix.

3.7 Unified Refinement Procedure

The complete methodology follows a consistent sequence from initialization to local projection correction and then to scalar interaction correction.

Stage 1: Calibration and Quantization

First, calibration activations are summarized into the empirical statistic $\tilde{x}$ and then shrunk toward target m to obtain the representative activation x . Next, each selected projection matrix is partitioned along the input dimension into quantization groups using the mapping γ . Group-specific asymmetric INT4 scales and zero points are computed from zero-inclusive ranges, and RTN initializes the stored integers. This stage produces dequantized query-, key-, and value-head approximations under a common legal integer representation while preserving the distinct GQA head-sharing structure.

Stage 2: Flip

For each projection selected for Flip, compute

$$E = \hat{W} - W, \quad r = xE^\top, \quad J = \|r\|_2^2. \quad (3.74)$$

Evaluate legal one-step integer candidates. For a candidate at (j, i) , use

$$\Delta J = 2r_j x_i \delta + (x_i \delta)^2, \quad \delta = \alpha_{j, \gamma(i)} \Delta z. \quad (3.75)$$

Accept only a strictly negative exact objective change and immediately update r_j . Repeat until the Flip stopping rule is met. The result is a projection matrix whose aggregate residual under x is no larger than at the beginning of the accepted sequence.

Stage 3: ReFlip

After the chosen Flip refinements, process every query head $h \in \mathcal{G}_r$ against its fixed shared key head r . Compute

$$s_{h,r}^* = \left[x(W_Q^{(h)})^\top \right] \cdot \left[x(W_K^{(r)})^\top \right], \quad \hat{s}_{h,r} = \left[x(\hat{W}_Q^{(h)})^\top \right] \cdot \left[x(\hat{W}_K^{(r)})^\top \right], \quad (3.76)$$

and set $e_{h,r} = \hat{s}_{h,r} - s_{h,r}^*$. Hold $\hat{W}_K^{(r)}$ fixed for every query head in $\mathcal{G}_r$. Compute integer-level and dimension-level sensitivities from current $\hat{k}^{(r)}$, apply the normalized decreasing-curve Kneedle configuration, include the detected knee in the protected region, and admit the moderate region immediately after it. If the curve has no valid knee, skip ReFlip for that query head.

For every allowed query candidate, evaluate

$$\Delta s_{ab}^{(h,r)} = \hat{k}_a^{(r)} x_b \alpha_{a,\gamma(b)}^{(h)} \Delta z_{ab}^{(h)} \quad (3.77)$$

and

$$\Delta \mathcal{L}_{ab}^{(h,r)} = e_{h,r} \Delta s_{ab}^{(h,r)} + \frac{1}{2} \left(\Delta s_{ab}^{(h,r)} \right)^2. \quad (3.78)$$

Accept only $\Delta \mathcal{L}_{ab}^{(h,r)} < 0$, update the per-head scalar residual after each accepted change, and terminate under the ReFlip stopping rule. No key or value integer is modified during this stage.

Monotonicity of Accepted Changes

The two stages have simple monotonicity properties under their own stated objectives. Every accepted Flip change satisfies

$$J_{\text{after}} < J_{\text{before}}, \quad (3.79)$$

and every accepted ReFlip change satisfies

$$\mathcal{L}_{\text{after}}^{(h,r)} < \mathcal{L}_{\text{before}}^{(h,r)}. \quad (3.80)$$

These properties follow from exact objective differences and immediate residual updates. They do not imply monotonic improvement in a different metric, such as unweighted weight reconstruction error, the complete attention matrix, language-model loss, or downstream task quality.

The order of the stages matters conceptually. Flip first improves each selected projection under an activation-weighted output objective. ReFlip then observes

each per-query-head surrogate residual left by the current quantized query and shared key heads and uses query-only compensation. Reversing the stages would change the scalar residual and sensitivity profile against which ReFlip operates.

3.8 Hyperparameters and Design Choices

The methodology exposes its principal choices as explicit hyperparameters.

- **Bit width B .** The present formulation fixes $B = 4$, giving the legal unsigned integer set $\mathcal{Z}_4 = \{0, \dots, 15\}$. The symbol B is distinct from input-column index b .
- **Quantization-group size G .** The group size determines the column-to-group mapping $\gamma(b)$ and the granularity of scales and zero points. Smaller groups increase quantization metadata but can better adapt to local weight ranges. This quantity is distinct from the GQA head set $\mathcal{G}_r$.
- **Calibration sample count N_{cal} .** This controls the amount of activation data used to estimate $\tilde{x}$, the shrinkage target, and the noise level.
- **James–Stein target m and noise estimate $\hat{\sigma}^2$.** These determine the shrinkage factor λ and therefore the representative activation used by both exact objectives.
- **Flip pass budget T_F .** This is the maximum number of scans over eligible Flip coordinates.
- **Flip acceptance budget B_F .** This limits the total number of accepted integer changes for a projection or layer.
- **Flip improvement tolerance ε_F .** In finite precision, the strict mathematical condition $\Delta J < 0$ may be implemented conservatively as $\Delta J < -\varepsilon_F$.
- **ReFlip Kneedle sensitivity κ .** This controls how the knee detector responds to curvature in the sorted dimension-sensitivity curve.
- **Moderate-region width H_R .** This determines how many sorted dimensions immediately after the knee are eligible for ReFlip.
- **ReFlip pass budget T_R and acceptance budget B_R .** These bound the greedy search over legal query-integer changes.
- **ReFlip improvement tolerance ε_R .** A conservative finite-precision rule accepts only $\Delta \mathcal{L}_{ab}^{(h,r)} < -\varepsilon_R$.
- **Candidate order.** Flip candidates may be ordered by potential activation-weighted step magnitude, while ReFlip candidates may be ordered by $\mathcal{S}_{ab}^{(h,r)}$.

Ordering affects the greedy path but never replaces the exact acceptance tests.

The one-step neighborhood $\Delta z \in \{-1, +1\}$ is part of the method rather than an unconstrained hyperparameter. It keeps changes local on the existing INT4 grid and makes both objective differences exact and inexpensive to evaluate. Multiple accepted one-step changes may eventually move an integer by more than one level if the candidate is revisited and every intermediate change remains legal and strictly beneficial.

Hyperparameter selection should respect the distinction between refinement opportunity and refinement risk. Very large budgets or a wide post-knee region create more opportunities to reduce the calibration objective but can increase dependence on the representative activation. Conservative budgets, shrinkage, protected extreme dimensions, and positive improvement tolerances are therefore used as design measures intended to limit such over-specialization; their protective effect is not presented as a formal guarantee.

3.9 Theoretical Discussion

From Scalar Reconstruction to Functional Residuals

The methodology is motivated by a hierarchy of objectives. RTN minimizes isolated scalar weight error for fixed quantization parameters. Flip moves to a projection-level functional residual,

$$x(\hat{W} - W)^\top, \quad (3.81)$$

which accounts for activation magnitude, error sign, and cancellation within each output row. For query head $h \in \mathcal{G}_r$, ReFlip then moves to the per-head scalar surrogate objective

$$\mathcal{L}_{h,r} = \frac{1}{2} \left[\hat{q}^{(h)} \cdot \hat{k}^{(r)} - q^{(h)} \cdot k^{(r)} \right]^2, \quad (3.82)$$

which accounts for the coupling between one query head and its shared GQA key head while constraining changes to the quantized query-head matrix.

Each level deliberately sacrifices breadth for tractability. The objectives are compact enough to permit exact evaluation of every legal one-step candidate. This avoids the ambiguity of claiming that a local scalar change optimizes a large downstream object through an unverified approximation. The trade-off is that exactness applies only to the specified representative objectives.

Why Query-Only Compensation Can Help

The surrogate residual can be expanded conceptually as the difference between a dot product of two quantized projected vectors and the corresponding full-precision dot product. Its value reflects errors in both vectors and their interaction. Once $\hat{W}_K^{(r)}$ is fixed, changing $\hat{W}_Q^{(h)}$ alters one side of the dot product while leaving the other side as the reference direction for correction. A suitable query-head change can therefore offset part of the surrogate error associated with the fixed shared quantized key head.

This compensation does not require the modified query weight to approach its original value. It requires only that the exact change in $\hat{s}_{h,r}$ reduce $|\hat{s}_{h,r} - s_{h,r}^*|$. Re-Flip is thus aligned with the same general principle as Flip: a functional aggregate can improve even when an individual scalar reconstruction becomes worse. The query projection residual can likewise become worse because it is not the ReFlip acceptance objective.

Role of Sensitivity Protection

Sensitivity motivates two competing design considerations. A high-sensitivity parameter can correct a residual with a small number of integer changes, but its larger score step can also overshoot the current scalar target. ReFlip therefore does not simply choose the largest sensitivities. As a conservative design choice intended to reduce dependence on extreme score steps and on the representative calibration vector, it protects dimensions in the extreme pre-knee region and searches the moderate region immediately after the knee.

This policy is intentionally risk-aware rather than optimal. The Kneedle partition adapts to the shape of each query head’s dimension-sensitivity distribution, while the exact loss test ensures that an admitted candidate is accepted only when it improves the current scalar objective. The separation between heuristic admission and exact acceptance is central: sensitivity controls where to search, whereas $\Delta\mathcal{L}_{ab}^{(h,r)}$ determines whether to change an integer.

Boundaries of the Claims

Several boundaries follow directly from the formulation. First, the representative activation is a James–Stein-shrunk calibration summary, not the complete activation distribution. Second, Flip exactly reduces only its projection residual objective for accepted changes. Third, ReFlip exactly reduces only its per-query-head representative-vector Q–K surrogate loss for accepted changes. Fourth, Re-Flip modifies $\hat{W}_Q^{(h)}$ only and makes no update to $\hat{W}_K^{(r)}$ or $\hat{W}_V^{(r)}$.

Most importantly, the ReFlip scalar surrogate does not preserve the full pairwise attention matrix. In a complete sequence, queries and keys arise from different token rows, and attention computation includes many cross-token dot products. A single unscaled surrogate based on x cannot constrain all of them. The method also does not directly optimize the softmax transformation or the value-weighted attention output. These limitations do not invalidate the targeted objectives, but they prevent stronger claims than the mathematics supports.

Discrete Search and Optimality

Both Flip and ReFlip perform greedy coordinate search on a finite integer grid. Their accepted changes are monotonic under their respective objectives, and termination is guaranteed under finite budgets. Without an explicit budget, strict decrease on a finite state space also rules out cycles, although repeated scans may be undesirable.

Neither procedure guarantees a globally optimal quantized matrix. Flip can miss coordinated changes that are beneficial only when applied together. ReFlip can miss a sequence whose first step is neutral or unfavorable but whose combined effect is favorable. Candidate ordering, budget choices, and the Kneedle-selected region can all influence the endpoint. Kneedle itself is a heuristic detector, not an optimality guarantee. The methodological claim is therefore limited to exact evaluation and strict improvement of accepted local changes under the stated objectives.

3.10 Summary

This chapter has defined a post-training refinement methodology for group-wise asymmetric INT4 attention projections under a consistent row-vector convention. A representative James–Stein-shrunk activation x provides the calibration signal. The initial RTN quantizer assigns legal integers z_{ab} with quantization-group-specific scales $\alpha_{a,\gamma(b)}$ and zero points $\zeta_{a,\gamma(b)}$. Zero-inclusive ranges define one-sided groups without ambiguity, while all-zero degenerate groups use an explicit exact frozen representation.

Flip refines a projection matrix through the exact objective

$$J = \|x(\hat{W} - W)^\top\|_2^2. \quad (3.83)$$

For a legal dequantized change δ at (j, i) , it evaluates

$$\Delta J = 2r_j x_i \delta + (x_i \delta)^2 \quad (3.84)$$

and accepts only strict reductions. RTN remains the minimizer of isolated scalar weight error; Flip can improve the aggregate activation-weighted residual by allowing signed error cancellation, even when an individual weight becomes less accurate.

For each query head $h \in \mathcal{G}_r$, ReFlip holds shared $\hat{W}_K^{(r)}$ fixed and updates $\hat{W}_Q^{(h)}$ only. It targets the unscaled representative-vector Q–K surrogate loss

$$\mathcal{L}_{h,r} = \frac{1}{2} \left[\hat{q}^{(h)} \cdot \hat{k}^{(r)} - q^{(h)} \cdot k^{(r)} \right]^2. \quad (3.85)$$

A legal unit integer change has exact surrogate effect

$$\Delta s_{ab}^{(h,r)} = \hat{k}_a^{(r)} x_b \alpha_{a,\gamma(b)}^{(h)} \Delta z_{ab}^{(h)}, \quad (3.86)$$

integer-level sensitivity

$$\mathcal{S}_{ab}^{(h,r)} = \left| \hat{k}_a^{(r)} x_b \alpha_{a,\gamma(b)}^{(h)} \right|, \quad (3.87)$$

and dimension sensitivity

$$\mathcal{S}_a^{(h,r)} = \left| \hat{k}_a^{(r)} \right| \left(\sum_{b \in \mathcal{B}_a^{(h)}} \left(\alpha_{a,\gamma(b)}^{(h)} \right)^2 x_b^2 \right)^{1/2}. \quad (3.88)$$

Kneedle is applied per query head to a unit-normalized decreasing, convex sensitivity curve. It includes the detected knee in the protected extreme region, admits a moderate region immediately after it, and conservatively skips a head when no valid knee exists. Each admitted candidate is evaluated using the exact loss change

$$\Delta \mathcal{L}_{ab}^{(h,r)} = e_{h,r} \Delta s_{ab}^{(h,r)} + \frac{1}{2} \left(\Delta s_{ab}^{(h,r)} \right)^2, \quad (3.89)$$

and the per-head scalar residual is updated after every accepted change. Such a change may worsen query projection residual while reducing the targeted scalar loss.

The resulting procedure retains the theoretical memory advantages and deployment form of group-wise INT4 weight-only quantization. Its guarantees are deliberately limited to the stated objectives. Kneedle is heuristic, greedy refinement is not globally optimal, and the scalar ReFlip surrogate does not preserve the full pairwise attention matrix. Within these boundaries, Flip first corrects the projection residual under the representative activation, and ReFlip then performs exact per-query-head compensation of the unscaled

representative-vector Q–K surrogate.

CHAPTER 4. EXPERIMENTAL EVALUATION

4.1 Experimental Setup

This chapter evaluates whether the proposed Flip and ReFlip refinements are accompanied by lower quantization distortion than standard round-to-nearest quantization and established post-training quantization baselines. The evaluation is organized into two complementary levels. The first is a tightly bounded standalone case study that follows the local refinement sequence and reports the scalar Q–K score distortion after RTN, Flip, and Flip + ReFlip for one grouped-query-attention setting. Flip’s projection-residual objective is part of this sequence, but the reported standalone tables focus on the scalar score surrogate used to compare the successive quantized states. The second level compares full-model perplexity and downstream task accuracy after 4-bit quantization. The local study directly examines the targeted representative-vector comparison, whereas the full-model comparison describes end-to-end behavior without isolating a single projection, layer, or correction decision.

The standalone case study follows the Transformer attention formulation [16] and the grouped-query-attention (GQA) structure used by modern decoder-only language models [5]. Its scope is deliberately narrow: one representative activation vector is evaluated at Llama-3-8B layer 8, GQA group 3, which contains four query heads sharing one key/value head. The representative vector is formed from C4 calibration activations [18]: 128 samples of sequence length 512 provide 65536 token-level activation observations over 4096 hidden dimensions, and a James–Stein-inspired shrinkage heuristic [7] regularizes the channel-wise mean used as the representative vector. The study evaluates the unscaled scalar Q–K surrogate for the four query heads. It does not evaluate an attention matrix, a complete attention block, or the value projection W_V . Table 4.1 summarizes this configuration and its evidence boundary.

Table 4.1: Configuration and evidence boundary of the standalone Q–K case study.

Item	Configuration
Evaluated setting	Llama-3-8B, layer 8, GQA group 3
GQA structure	Four query heads sharing one key/value head
Calibration observations	128 C4 samples, sequence length 512; 65536 $\times$ 4096 activations
Evaluation input	One James–Stein-shrunk representative activation vector
Weight quantization	Asymmetric INT4, group size 128
Evaluated projections	W_Q and W_K
Evaluated quantity	Unscaled scalar Q–K surrogate per query head
Not evaluated	W_V , attention matrices, and complete attention outputs

The full-model comparison covers Mistral-7B [19], Llama-3/Llama-3.1 8B [20], and Qwen2.5-7B [21]. Only the metrics considered in this thesis are reported: WikiText-2 perplexity, C4 perplexity, ARC-Challenge, ARC-Easy, BoolQ, PIQA, RTE, and the average of the five accuracy tasks. WikiText-2 is a language modeling benchmark introduced with the WikiText corpus [22], while C4 originates from the T5 work [18]. ARC-Challenge and ARC-Easy measure science-question reasoning [23]; BoolQ measures natural yes/no question answering [24]; PIQA measures physical commonsense reasoning [25]; and RTE is included through the GLUE benchmark family [26]. A reproducible language-model evaluation framework supports consistent task definitions across models [27]. Only reported metric values are compared; unavailable values are neither inferred nor substituted.

For WikiText-2 and C4, lower perplexity is better. For ARC-Challenge, ARC-Easy, BoolQ, PIQA, RTE, and their average, higher accuracy is better. The reported average is computed over the five selected downstream accuracy metrics:

$$\text{avg} = \frac{\text{arc-c} + \text{arc-e} + \text{boolq} + \text{pqa} + \text{rte}}{5}.$$

Full precision is shown as a reference baseline rather than as a competitor in the quantized-method ranking. It is not assumed to be an absolute ceiling because a finite evaluation can occasionally favor a quantized model on an individual metric. In the full-model tables, bold values indicate the best reported quantized result for a metric within the same model, and underlined values indicate the second-best reported quantized result. Ties at the second-best value are all underlined. Full-precision values are not highlighted.

The method names are normalized for clarity. The full-precision model is denoted as *Full precision*. Standard round-to-nearest quantization is denoted as *RTN*; it is the direct rounding baseline commonly used before more data-aware reconstruction is considered [1], [11], [12]. GPTQ uses approximate second-order information to quantize pretrained Transformer weights [13]. AWQ is an activation-aware weight-only quantization method that protects activation-salient feature channels using calibration statistics [3]. A method labelled “+ Flip” applies the Flip refinement to its named baseline, while “+ ReFlip” denotes the corresponding result after the ReFlip refinement. In the standalone case study, where ReFlip explicitly begins from the Flip-refined state, the complete third stage is labelled *Flip + ReFlip*.

These two evaluation levels answer different questions. The standalone case study measures the exact quantity selected for its local comparison, so it can show how the three successive quantized states differ for the representative vector. The full-model tables instead report aggregate behavioral metrics influenced by all model components. They can establish which reported model–method combinations have lower perplexity or higher task accuracy, but they cannot by themselves attribute a full-model difference exclusively to a particular local correction.

4.2 Standalone Q–K Surrogate Evaluation

The standalone case study compares three successive quantized states under the same INT4 group-wise quantization setting. RTN performs direct round-to-nearest quantization. Flip then adjusts selected integer levels to reduce the aggregate activation-weighted projection residual. Finally, ReFlip begins from the Flip-refined state, keeps the quantized key projection fixed, and updates selected query-weight integer levels according to the remaining scalar Q–K surrogate loss. The resulting third state is therefore denoted as *Flip + ReFlip*, rather than ReFlip alone.

For query head h , let s_h^* be the full-precision scalar Q–K surrogate and let $\hat{s}_{h,m}$ be the score obtained by quantized state m . The reported absolute and relative errors are

$$e_{h,m} = |\hat{s}_{h,m} - s_h^*|, \quad e_{h,m}^{\text{rel}} = \frac{e_{h,m}}{|s_h^*|} \times 100\%.$$

Across the four query heads, mean absolute error (MAE) and maximum absolute error are

$$\text{MeanRel}_m = \frac{1}{4} \sum_{h=1}^4 e_{h,m}^{\text{rel}}, \quad \text{MAE}_m = \frac{1}{4} \sum_{h=1}^4 e_{h,m}, \quad e_m^{\text{max}} = \max_h e_{h,m}.$$

Two reduction summaries are reported because they weight heads differently. For a baseline state b and a later state m , the mean of per-head percentage reductions gives equal importance to every head after normalizing by its own baseline error:

$$R_{\text{head}}(b \rightarrow m) = \frac{100}{4} \sum_{h=1}^4 \frac{e_{h,b} - e_{h,m}}{e_{h,b}}.$$

The aggregate-MAE reduction first aggregates absolute errors and then computes one percentage:

$$R_{\text{MAE}}(b \rightarrow m) = 100 \frac{\text{MAE}_b - \text{MAE}_m}{\text{MAE}_b}.$$

Consequently, R_{head} gives a relatively small-error head the same influence as a large-error head, whereas R_{MAE} places more influence on heads contributing more absolute error.

Table 4.2: Standalone scalar Q–K surrogate error on Llama-3-8B layer 8, GQA group 3.

Method	Mean abs. error	Max abs. error	Mean rel. error (%)	R_{head} (%)	R_{MAE} (%)	Incremental operations	Cumulative operations
RTN	0.142806	0.253523	0.5330	–	–	0	0
Flip	0.070070	0.113184	0.2666	41.63	50.93	20,889	20,889
Flip + ReFlip	0.047939	0.068937	0.1934	56.63	66.43	64	20,953

Table 4.2 separates the one-level integer operations accepted at each stage from the cumulative number of accepted operations across the sequence. Flip accepts 20,889 operations across the selected Q and K projections, corresponding to approximately 0.7969% of the involved weights. ReFlip then accepts 64 additional query-weight operations, bringing the cumulative operation count to 20,953. This sum is an operation count, not necessarily the number of distinct final weights that differ from RTN, because a coordinate may in principle be revisited. The additional stage is accompanied by a decrease in mean absolute scalar-score error from 0.070070 to 0.047939. Relative to RTN, the final state has a 56.63% mean per-head reduction and a 66.43% aggregate-MAE reduction. Relative to Flip itself, the corresponding incremental reductions are 27.14% and 31.58%, respectively.

Table 4.3: Per-head scalar Q–K surrogate errors in the standalone case study.

Head	RTN error	Flip error	Flip + ReFlip error	RTN to Flip (%)	RTN to Flip + ReFlip (%)
0	0.092831	0.062511	0.034649	32.66	62.68
1	0.253523	0.113184	0.068937	55.36	72.81
2	0.192633	0.078560	0.063698	59.22	66.93
3	0.032239	0.026023	0.024472	19.28	24.09

The per-head results in Table 4.3 show lower absolute scalar-score error for

the Flip + ReFlip state than for RTN or Flip on all four query heads in this case study. The magnitude is not uniform across heads: head 1 has the largest RTN-to-Flip + ReFlip percentage reduction, while head 3 has the smallest. The arithmetic distinction between the two reduction summaries is visible here. Head 3 has the smallest RTN absolute error and therefore contributes little to aggregate MAE, yet its comparatively modest percentage reduction receives equal weight in R_{head} . This is why the 56.63% mean per-head reduction is lower than the 66.43% aggregate-MAE reduction.

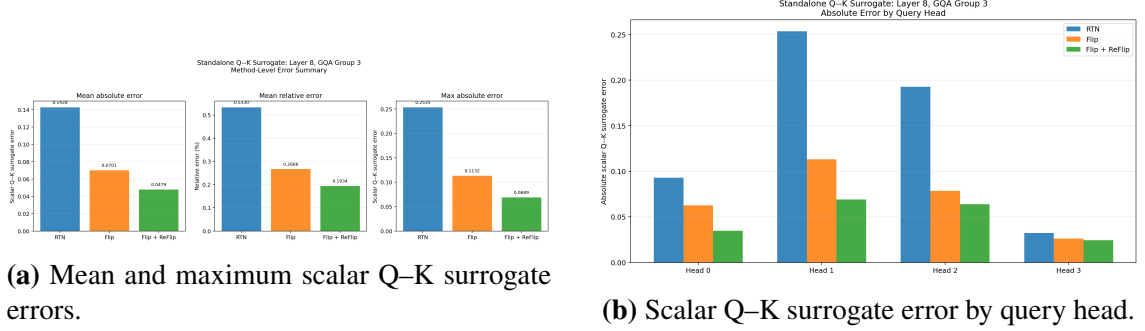


Figure 4.1: Standalone scalar Q-K surrogate error comparison across RTN, Flip, and Flip + ReFlip.

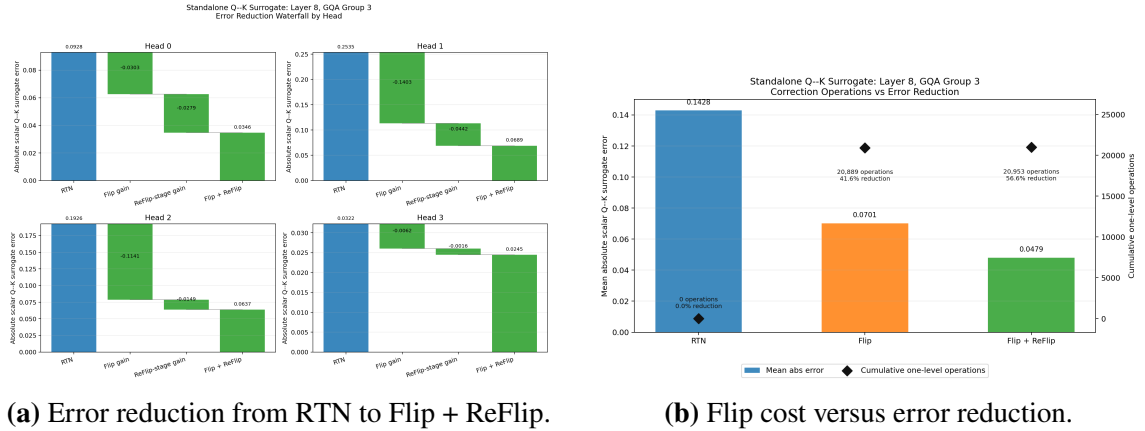


Figure 4.2: Per-head scalar Q-K surrogate correction behavior and cost-benefit analysis.

Figures 4.1 and 4.2 visualize the same bounded case study. The bars compare independent query heads; adjacency on the horizontal axis does not imply an ordering relationship or a trend between heads. Flip is associated with the larger first decrease in absolute scalar-score error and accepts operations affecting a small fraction of the selected projection weights. The subsequent ReFlip stage is associated with a further decrease while introducing 64 additional one-level operations, so within this representative-vector comparison it behaves as a targeted correction rather than a broad re-quantization pass.

The evidence boundary is important. These numbers characterize one representative activation vector at one layer and one GQA group. They do not measure token-to-token attention matrices, softmax-normalized attention probabilities, value-projection distortion, or final attention outputs. In particular, W_V is not evaluated in this standalone study. The results therefore demonstrate the behavior of the selected scalar Q–K surrogate in this case, but they do not establish that every activation, head group, layer, or complete attention output changes by the same proportion.

4.3 Full-Model Evaluation

The full-model results describe end-to-end behavior for four model configurations: Mistral-7B, Llama-3.1-8B, Qwen2.5-7B, and Llama-3-8B. Tables 4.4, 4.5, 4.6, and 4.7 report the available 4-bit quantization results for RTN, AWQ, GPTQ, and their Flip/ReFlip variants. Perplexity and downstream accuracy reflect the complete model and therefore complement, rather than directly validate, the isolated scalar Q–K case study. A lower perplexity indicates that the evaluated model assigns higher probability to the observed language sequence on average. Higher downstream accuracy indicates a larger proportion of correct task predictions. The five-task average gives each selected accuracy task equal weight, regardless of dataset size.

The tables are interpreted descriptively. A difference between a baseline and its Flip or ReFlip variant is an observed association for the reported model and metric; it does not isolate which layer, projection, or individual correction produced the difference. Complete repeated-run metadata, uncertainty estimates, and every protocol detail—including all calibration, context-length, prompting, shot-setting, and random-seed choices—are not available for every configuration. The comparisons therefore rely on the reported configurations being internally consistent and should not be interpreted as statistical significance tests or as evidence about run-to-run variance. No unavailable metric is filled or inferred. Bold and underline identify the first- and second-ranked quantized values within each model after excluding the full-precision reference baseline.

Table 4.4: Full-model 4-bit evaluation for Mistral-7B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	4.845	7.535	0.509	0.802	0.837	0.800	0.686	0.7268
RTN	4.9917	7.8047	0.4761	0.7875	<u>0.8248</u>	0.7992	0.6643	0.7104
RTN + ReFlip	4.9704	<u>7.7167</u>	0.4804	0.7934	0.8202	<u>0.8014</u>	0.6751	0.7141
RTN + Flip	4.9948	7.738	0.4744	0.7887	0.8275	0.7987	0.6606	0.7100
AWQ	4.975	7.727	<u>0.493</u>	0.794	0.817	0.797	0.671	0.7144
AWQ + ReFlip	4.965	7.711	0.495	<u>0.795</u>	0.818	0.802	0.690	0.7200
AWQ + Flip	4.975	7.726	<u>0.493</u>	0.793	0.815	0.798	<u>0.678</u>	0.7154
GPTQ	4.978	7.744	0.480	<u>0.795</u>	0.820	0.798	0.675	0.7136
GPTQ + ReFlip	<u>4.967</u>	7.736	0.483	0.796	0.821	0.800	<u>0.678</u>	<u>0.7156</u>
GPTQ + Flip	4.979	7.745	0.480	<u>0.795</u>	0.819	0.798	0.675	0.7134

Table 4.5: Full-model 4-bit evaluation for Llama-3.1-8B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	5.522	8.409	0.522	0.822	0.830	0.791	0.711	0.7352
RTN	6.0248	9.4397	0.4898	0.8064	<u>0.7990</u>	0.7835	0.6351	0.7028
RTN + ReFlip	5.9727	9.3692	0.5051	<u>0.8068</u>	0.7928	0.7867	0.6531	0.7089
RTN + Flip	6.0815	9.6308	0.4872	0.8009	0.8075	0.7845	0.6315	0.7023
AWQ	<u>5.968</u>	9.321	<u>0.511</u>	0.806	0.794	0.798	0.635	0.7088
AWQ + ReFlip	5.949	9.294	0.512	0.808	0.796	0.796	0.671	0.7166
AWQ + Flip	<u>5.968</u>	9.316	0.507	0.803	0.790	0.796	0.632	0.7056
GPTQ	5.995	9.304	0.505	0.804	0.795	0.793	0.651	0.7096
GPTQ + ReFlip	5.989	<u>9.300</u>	0.507	0.806	0.790	<u>0.797</u>	<u>0.663</u>	<u>0.7126</u>
GPTQ + Flip	5.992	9.304	0.499	0.803	0.792	0.793	0.659	0.7092

Table 4.6: Full-model 4-bit evaluation for Qwen2.5-7B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	6.0839	10.8166	0.477	0.804	0.846	0.787	0.815	0.7458
RTN	6.6915	12.2029	0.4506	0.7784	0.8302	0.7662	0.8016	0.7254
RTN + ReFlip	6.6366	11.8779	0.45162	0.77125	0.83264	0.76782	0.816018	0.7278696
RTN + Flip	6.6427	12.1076	0.44138	0.76832	0.82685	0.7651	0.823258	0.7249816
AWQ	6.4677	<u>11.4443</u>	<u>0.469</u>	0.786	0.842	0.780	0.805	0.7364
AWQ + ReFlip	6.438	11.363	0.473	0.792	0.847	<u>0.782</u>	<u>0.819</u>	0.7426
AWQ + Flip	<u>6.467</u>	11.450	0.459	<u>0.787</u>	<u>0.843</u>	0.787	0.815	<u>0.7382</u>
GPTQ	6.708	11.539	0.454	0.781	0.805	0.780	0.769	0.7178
GPTQ + ReFlip	6.702	11.534	0.459	0.779	0.814	0.779	0.787	0.7236
GPTQ + Flip	6.709	11.538	0.451	0.779	0.805	0.778	0.769	0.7164

Table 4.7: Full-model 4-bit evaluation for Llama-3-8B. Bold marks the best quantized result; underline marks the second-best quantized result.

Method	Wiki	C4	arc-c	arc-e	boolq	piqa	rte	avg
Full precision	5.442	8.346	0.505	0.800	0.813	0.796	0.696	0.7220
RTN	6.0572	9.5846	0.4789	0.780	0.791	0.784	0.628	0.6924
RTN + ReFlip	6.0236	9.2523	<u>0.490</u>	0.781	0.796	0.7818	0.6679	0.7033
RTN + Flip	6.0711	9.3279	0.4806	0.782	0.794	0.7802	0.6390	0.6952
AWQ	5.910	9.240	0.488	0.784	0.795	<u>0.791</u>	0.646	0.7008
AWQ + ReFlip	5.885	9.199	0.493	<u>0.788</u>	0.799	0.788	0.679	<u>0.7094</u>
AWQ + Flip	<u>5.906</u>	<u>9.234</u>	0.489	0.784	0.796	0.792	0.617	0.6956
GPTQ	6.011	9.370	0.487	0.783	<u>0.798</u>	0.788	<u>0.689</u>	0.7090
GPTQ + ReFlip	6.013	9.354	0.489	0.790	0.795	0.788	0.693	0.7110
GPTQ + Flip	6.014	9.371	0.486	0.783	0.795	0.787	0.686	0.7074

4.4 Discussion

The standalone case study provides the most direct comparison with the local scalar surrogate considered in this thesis. For the selected representative activation vector, quantization errors in Q and K combine through their dot product. The reported states show a decrease in mean absolute scalar-score error from 0.142806 under RTN to 0.070070 after Flip and to 0.047939 after Flip + ReFlip. However, the numerical size of a percentage reduction depends on the aggregation rule. Equal weighting of the four per-head percentages gives reductions of 41.63% for Flip and 56.63% for Flip + ReFlip relative to RTN. Computing the percentage after aggregating absolute errors gives larger reductions of 50.93% and 66.43%, respectively, because the heads with larger RTN errors also receive larger absolute corrections.

Reporting both prevents the smaller-error heads from being hidden by the aggregate while retaining a direct account of total absolute distortion.

The correction-operation counts provide a second perspective on the same local comparison. Flip accounts for 20,889 accepted one-level operations in the selected Q and K projections. The ReFlip stage contributes 64 additional query-weight operations, bringing the cumulative operation count to 20,953. The final error is therefore observed after the complete sequence of Flip followed by ReFlip; neither the 64 operations nor their cumulative sum should be interpreted as the number of distinct weights separating RTN from the final state without a separate final-state comparison. Within this case study, the additional 64 operations coincide with a 27.14% mean per-head reduction, or a 31.58% aggregate-MAE reduction, relative to the Flip state. These observations are consistent with ReFlip acting as a narrower second-stage adjustment for the selected scalar objective, but they do not establish the same cost–benefit relationship outside this layer, group, and representative vector.

The full-model task results are more heterogeneous because each metric depends on the complete network rather than on one scalar score in one GQA group. On Mistral-7B, AWQ + ReFlip has the best quantized WikiText-2 and C4 perplexities, ARC-Challenge, PIQA, RTE, and five-task average. GPTQ + ReFlip has the highest quantized ARC-Easy score, while RTN + Flip has the highest quantized BoolQ score. The best quantized method therefore varies by task even though AWQ + ReFlip leads the average. Relative to their named baselines, all three + ReFlip variants have a higher five-task average: RTN changes from 0.7104 to 0.7141, AWQ from 0.7144 to 0.7200, and GPTQ from 0.7136 to 0.7156.

On Llama-3.1-8B, AWQ + ReFlip is the best quantized method for both perplexity metrics, ARC-Challenge, ARC-Easy, RTE, and the five-task average of 0.7166. GPTQ + ReFlip is second on the average with 0.7126. The recalculated RTN-family averages are 0.7028 for RTN, 0.7089 for RTN + ReFlip, and 0.7023 for RTN + Flip. Thus, RTN + ReFlip is observed above RTN on the selected-task average and below it on both perplexity measures, while RTN + Flip has the highest quantized BoolQ score but a slightly lower average than RTN. This contrast illustrates why an isolated winning task score and an across-task average should be interpreted separately.

On Qwen2.5-7B, AWQ + ReFlip is the best quantized method for WikiText-2, C4, ARC-Challenge, ARC-Easy, BoolQ, and the selected-task average. AWQ + Flip is second on the average and has the best quantized PIQA score. RTN + Flip

has the best quantized RTE score, whereas RTN + ReFlip has the higher RTN-family average, changing from 0.7254 for RTN to 0.7279. These reported values again show that the ranking of a refinement can depend on whether the comparison concerns one task, perplexity, or an unweighted task average.

On Llama-3-8B, AWQ + ReFlip has the best quantized WikiText-2 and C4 perplexity and the highest quantized ARC-Challenge and BoolQ scores. GPTQ + ReFlip has the highest quantized ARC-Easy, RTE, and average score. The named-baseline comparisons for average accuracy are 0.6924 to 0.7033 for RTN and RTN + ReFlip, 0.7008 to 0.7094 for AWQ and AWQ + ReFlip, and 0.7090 to 0.7110 for GPTQ and GPTQ + ReFlip. In this model, the highest average and the lowest perplexity are therefore obtained by different quantized method families.

Across the four models, every reported + ReFlip variant has a higher five-task average than its corresponding RTN, AWQ, or GPTQ baseline. The + ReFlip variants also have lower perplexity than their named baselines in all reported pairings except the Llama-3-8B GPTQ WikiText-2 comparison, where 6.013 is slightly higher than 6.011. By contrast, the + Flip averages are more variable: some exceed their named baseline and others do not. These are descriptive regularities in the reported comparisons, not proof that one refinement causes a universal improvement or that the same ranking will hold for unreported tasks, models, quantization settings, or random variations.

Evidence Boundaries and Limitations

Several limitations constrain the conclusions that can be drawn. First, the standalone evaluation uses one representative activation vector rather than a distribution of token-dependent query and key activations. Its scalar Q–K surrogate does not include the usual attention scaling, token-to-token score matrix, causal mask, softmax normalization, or value aggregation. It therefore measures a specific local distortion, not complete attention behavior. Second, only one layer, one GQA group, and four query heads are examined in the standalone case study. Variation across layers, groups, and activation contexts remains outside its scope.

Third, the standalone comparison does not evaluate W_V . Consequently, it cannot measure whether a lower scalar Q–K surrogate error produces a lower final attention-output error after value mixing. Fourth, the full-model tables aggregate effects from the entire network. They do not expose which local changes contribute to a perplexity or task-accuracy difference, and they do not constitute an ablation that isolates Flip or ReFlip from every other source of variation. Finally, task accuracy is discrete and benchmark-specific, while the five-task average gives each

selected task equal weight. A higher average can coexist with a lower score on an individual task, as several tables demonstrate.

Within these boundaries, the two evaluation levels remain complementary. The standalone results describe how RTN, Flip, and Flip + ReFlip differ for the exact representative-vector scalar surrogate under examination. The full-model results describe the perplexity and selected-task accuracy associated with the reported 4-bit model variants. Taken together, they support a cautious conclusion: the refinements are associated with lower local scalar-score distortion in the bounded case study and with frequently improved full-model metrics, while neither level establishes a universal guarantee across all attention states or downstream tasks.

CHAPTER 5. CONCLUSIONS

5.1 Summary

This thesis studied post-training weight quantization for large language models, with particular attention to the query, key, and value projections within grouped-query attention. Low-bit weight-only quantization can substantially reduce model storage and the memory traffic associated with inference. However, replacing full-precision weights with values on a coarse integer grid also perturbs the projection responses that participate in attention. The resulting distortion is not determined solely by the reconstruction error of each individual weight: it also depends on the activation multiplying that weight and on the interaction between the query and key projections.

To address these coupled effects, the thesis formulated two distinct refinement stages, Flip and ReFlip. Flip is an aggregate residual correction applied after an initial quantized state, including round-to-nearest in the standalone sequence. It revisits admissible integer levels and accepts a one-level change only when that change reduces the activation-weighted projection residual. Consequently, Flip does not require every modified weight to become individually closer to its full-precision value. A local reconstruction error may increase while the weighted errors within a projection row cancel more effectively, producing a smaller aggregate response error.

ReFlip addresses a different objective. It holds the quantized key projection fixed and updates only the quantized query projection to reduce a scalar query-key score surrogate. This query-only restriction makes the correction direction explicit and separates the score-level objective from the projection-level objective used by Flip. Parameter-level score sensitivities are aggregated to identify query dimensions with moderate correction potential, while Kneedle is used only to delimit a candidate region. Every proposed integer-level change must still be assessed against its exact change in the scalar surrogate loss. Thus, neither the candidate selection rule nor the greedy sequence of accepted changes is presented as a guarantee of global optimality.

The experimental evaluation considered both a narrowly controlled standalone setting and broader model-level metrics. The standalone analysis isolates the local quantity targeted by ReFlip and makes the behavior of the two refinement stages observable. The broader evaluation describes how quantized model variants behave on language-modeling and downstream tasks. Together, these two perspectives pro-

vide complementary evidence: the first examines the intended local correction mechanism, while the second indicates whether the resulting quantized variants remain competitive at model level. The evidence is interpreted descriptively and within the boundaries of the evaluated settings.

5.2 Main Findings

The standalone case study provides the most direct evidence for the local mechanisms examined in this thesis. By construction, accepted Flip changes reduce the activation-weighted projection objective under the representative vector. The reported standalone score table further shows smaller scalar query–key score mismatches after Flip and after the subsequent Flip + ReFlip state. This observed ordering is consistent with the objectives assigned to the two stages: Flip operates on an aggregate projection residual, whereas ReFlip operates on the residual of a scalar bilinear surrogate. The result does not establish that either stage minimizes a general attention loss, but it shows that the selected discrete corrections can be aligned with their respective local objectives.

The case study also illustrates why Flip and ReFlip should not be treated as interchangeable names for the same operation. Flip evaluates the effect of a candidate integer change through the projection residual induced by a representative activation vector. ReFlip begins from the Flip-refined state, fixes the key projection, and evaluates query-weight candidates through their exact effect on the current scalar score residual. A correction that is attractive under one objective is not necessarily attractive under the other. In particular, a ReFlip update may reduce the scalar surrogate loss while increasing the query projection residual. The observed improvements must therefore be understood as objective-specific rather than as uniform improvements of every reconstruction criterion.

The accepted corrections are sparse relative to a complete re-quantization of the selected projections. This behavior is compatible with the intended role of both methods as post-quantization refinements: they preserve the established quantization grid and alter only admissible integer levels whose evaluated loss changes are favorable. Sparsity alone does not establish efficiency on a particular hardware platform, but it indicates that the correction mechanism does not require replacing the quantization representation or optimizing every weight jointly.

At model level, the evaluated quantized variants display task-dependent differences in perplexity and downstream accuracy. Refined variants are competitive in several comparisons, while the relative ordering varies across models, baselines, and metrics. These results are best interpreted as descriptive evidence that the lo-

cal corrections can coexist with multiple quantization baselines. They do not, by themselves, isolate the contribution of every design choice or demonstrate that a reduced scalar score mismatch necessarily causes an improvement on a downstream task. Such a causal conclusion would require broader controlled ablations and repeated evaluations.

Taken together, the findings support a careful conclusion. Aggregate activation-weighted residuals and scalar query–key score residuals provide meaningful alternatives to individual weight reconstruction error when selecting discrete post-quantization corrections. Flip and ReFlip offer two complementary ways to exploit these objectives. Their usefulness, however, remains conditional on the chosen representative statistics, candidate-selection policy, quantization configuration, and evaluation setting.

5.3 Limitations

The first limitation concerns the representative activation vector. Both the activation-weighted Flip objective and the scalar ReFlip surrogate summarize a broader activation distribution through a single representative vector. This makes the discrete loss changes tractable and interpretable, but it cannot preserve every token-specific or sequence-specific projection response. A correction that is favorable for the representative vector may have a smaller benefit, no benefit, or an adverse effect for other activations. The conclusions therefore apply to the evaluated representative objective rather than to all possible inputs.

Second, the scalar query–key score surrogate is not the full attention matrix. It considers an unscaled bilinear interaction for a selected query head and its associated key head under grouped-query attention. It does not model all token-to-token logits, softmax normalization, causal masking, interactions among sequence positions, or the subsequent multiplication by value representations. Reducing this scalar mismatch is a focused local objective; it is not equivalent to preserving the complete attention operation.

Third, the optimization procedures are local and greedy. Kneedle identifies a region of moderate dimension-level sensitivity, but a knee in the sorted sensitivity curve does not certify that the selected region contains the globally best corrections. Within that region, integer-level candidates are accepted sequentially according to their current exact local loss change. Earlier accepted changes alter the residual used to evaluate later candidates, and the resulting path may differ from a joint optimum. Neither Kneedle selection nor greedy acceptance provides a global optimality guarantee.

Fourth, the standalone analysis has a deliberately narrow scope. It examines one selected attention layer, one grouped-query attention group, a small collection of query heads, and a single representative activation statistic. This scope is valuable for inspecting the correction mechanism without conflating it with the many transformations in a complete model, but it limits generalization across layers, head groups, architectures, calibration distributions, and quantization settings. Broader repeated studies are required before drawing claims about consistent behavior throughout an entire model.

Finally, the selected model-level metrics cover only part of the deployment problem. Perplexity and downstream accuracy do not directly characterize long-context behavior, generation quality, robustness, latency, memory bandwidth, energy consumption, or hardware-specific inference throughput. Likewise, the present comparisons do not constitute a complete ablation of the representative statistic, sensitivity aggregation, candidate budget, or knee-selection policy. These dimensions remain necessary for a more comprehensive assessment.

5.4 Future Work

Future evaluation should extend both the controlled and model-level settings. Repeating the standalone analysis across layers, GQA groups, query heads, calibration samples, and model families would reveal whether the observed correction patterns are stable or localized. A larger task suite, long-context benchmarks, repeated trials, and deployment measurements would provide a stronger account of quality, variance, memory behavior, and inference efficiency. Controlled ablations should separately examine the representative-vector estimator, the sensitivity aggregation rule, the Kneedle configuration, and the correction budget.

The correction strategy can also be generalized. ReFlip currently uses exact scalar-surrogate loss changes inside a heuristically selected candidate region and modifies only the query projection while holding the key projection fixed. Future work may study alternating query and key corrections, group-level objectives that aggregate several query heads sharing one key head, and objectives defined over multiple representative activations. Joint or beam-search procedures could also be compared with greedy acceptance to measure the quality lost through local decisions. These extensions must retain explicit integer constraints so that an apparent continuous improvement corresponds to a valid change on the quantization grid.

A natural direction is to formulate a value-projection correction through a fixed-attention value-output objective rather than through a query-key score. This objective evaluates the output after value mixing while deliberately holding

the attention-probability matrix fixed; it therefore targets the value path without implying end-to-end optimization of the complete attention operation. Let $X \in \mathbb{R}^{n \times d_{\text{in}}}$ be a matrix of representative activation rows, let $W_V, \hat{W}_V \in \mathbb{R}^{d_v \times d_{\text{in}}}$, and let $\bar{P} \in \mathbb{R}^{n \times n}$ be a fixed row-normalized attention-probability matrix. Define

$$O_V^* = \bar{P}V^*, \quad \hat{O}_V = \bar{P}\hat{V}, \quad L_V = \frac{1}{2} \left\| \hat{O}_V - O_V^* \right\|_F^2,$$

where $V^* = XW_V^\top \in \mathbb{R}^{n \times d_v}$ and $\hat{V} = X\hat{W}_V^\top \in \mathbb{R}^{n \times d_v}$. Using the same fixed $\bar{P}$ in both O_V^* and $\hat{O}_V$ makes the objective value-path specific: it does not ask the value projection to compensate for a mismatch between different attention matrices.

For a proposed one-level integer change $\Delta z_{ab} \in \{-1, +1\}$ in a quantized value weight, the corresponding dequantized weight change is

$$\Delta \hat{W}_{V,ab} = \alpha_{a,\gamma(b)} \Delta z_{ab}.$$

The following integer-level loss-change derivation is exact under the local correction setting in which $\bar{P}$, the quantization scale $\alpha_{a,\gamma(b)}$, the associated zero-point, and every weight other than the selected value weight are held fixed throughout evaluation of the candidate. This produces a structured value-response change $\Delta \hat{V}$ and an output change

$$\Delta \hat{V} = X \left(\Delta \hat{W}_V \right)^\top = \alpha_{a,\gamma(b)} \Delta z_{ab} (X e_b) e_a^\top, \quad \Delta \hat{O}_V = \bar{P} \Delta \hat{V}.$$

Here, e_a and e_b denote standard basis vectors for the output and input dimensions, respectively. The rank-one form makes the effect of a legal integer move explicit before it is propagated through the fixed attention matrix. Writing the current output residual as $R_O = \hat{O}_V - O_V^*$, the exact loss change for that admissible integer move is

$$\Delta L_V = \left\langle R_O, \Delta \hat{O}_V \right\rangle_F + \frac{1}{2} \left\| \Delta \hat{O}_V \right\|_F^2.$$

An output-aware value correction could accept the candidate only when $\Delta L_V < 0$, update R_O , and then evaluate the next candidate against the new residual. This formulation preserves the discrete, exact-loss-change principle used by Flip and ReFlip while aligning the value-weight objective with the representation produced after attention mixing. As with the query-only formulation, a greedy sequence of value corrections would remain a local procedure rather than a guarantee of global optimality.

More broadly, future work should investigate how projection-level, score-level,

and output-level objectives can be combined without obscuring their distinct meanings. A unified method could allocate correction budgets according to the residual that each projection can directly influence, while retaining transparent acceptance criteria for every integer-level adjustment. Such a formulation would provide a principled path from local quantization-error correction toward preservation of the complete attention computation.

REFERENCE

- [1] A. Gholami et al., “A survey of quantization methods for efficient neural network inference,” *arXiv preprint arXiv:2103.13630*, 2021.
- [2] M. Nagel et al., “A white paper on neural network quantization,” *arXiv preprint arXiv:2106.08295*, 2021.
- [3] J. Lin et al., “AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration,” in *Proceedings of Machine Learning and Systems*, vol. 6, 2024.
- [4] N. Shazeer, “Fast transformer decoding: One write-head is all you need,” *arXiv preprint arXiv:1911.02150*, 2019.
- [5] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebron, and S. Sanghai, “GQA: Training generalized multi-query transformer models from multi-head checkpoints,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, 2023, pp. 4895–4901.
- [6] V. Satopaa, J. Albrecht, D. Irwin, and B. Raghavan, “Finding a kneedle in a haystack: Detecting knee points in system behavior,” in *2011 31st International Conference on Distributed Computing Systems Workshops*, 2011, pp. 166–171. DOI: 10.1109/ICDCSW.2011.20
- [7] W. James and C. Stein, “Estimation with quadratic loss,” in *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, University of California Press, 1961, pp. 361–379.
- [8] H. Touvron et al., “LLaMA: Open and efficient foundation language models,” *arXiv preprint arXiv:2302.13971*, 2023.
- [9] J. Bai et al., “Qwen technical report,” *arXiv preprint arXiv:2309.16609*, 2023.
- [10] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han, “SmoothQuant: Accurate and efficient post-training quantization for large language models,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 38 087–38 099.
- [11] B. Jacob et al., “Quantization and training of neural networks for efficient integer-arithmetic-only inference,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 2704–2713.
- [12] M. Nagel, R. A. Amjad, M. Van Baalen, C. Louizos, and T. Blankevoort, “Up or down? adaptive rounding for post-training quantization,” in *Proceedings*

- of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 7197–7206.
- [13] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, “GPTQ: Accurate post-training quantization for generative pre-trained transformers,” in *International Conference on Learning Representations*, 2023.
 - [14] C. Hooper et al., “KVQuant: Towards 10 million context length LLM inference with KV cache quantization,” *arXiv preprint arXiv:2401.18079*, 2024.
 - [15] S. Dong, W. Cheng, J. Qin, and W. Wang, “QAQ: Quality adaptive quantization for LLM KV cache,” *arXiv preprint arXiv:2403.04643*, 2024.
 - [16] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.
 - [17] R. Banner, Y. Nahshan, E. Hoffer, and D. Soudry, “Post-training 4-bit quantization of convolutional networks for rapid-deployment,” in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
 - [18] C. Raffel et al., “Exploring the limits of transfer learning with a unified text-to-text transformer,” *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.
 - [19] A. Q. Jiang et al., “Mistral 7B,” *arXiv preprint arXiv:2310.06825*, 2023.
 - [20] A. Grattafiori et al., “The Llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.
 - [21] A. Yang et al., “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024.
 - [22] S. Merity, C. Xiong, J. Bradbury, and R. Socher, “Pointer sentinel mixture models,” *arXiv preprint arXiv:1609.07843*, 2016.
 - [23] P. Clark et al., “Think you have solved question answering? try arc, the ai2 reasoning challenge,” *arXiv preprint arXiv:1803.05457*, 2018.
 - [24] C. Clark, K. Lee, M.-W. Chang, T. Kwiatkowski, M. Collins, and K. Toutanova, “BoolQ: Exploring the surprising difficulty of natural yes/no questions,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, Association for Computational Linguistics, 2019, pp. 2924–2936. DOI: 10.18653/v1/N19-1300
 - [25] Y. Bisk, R. Zellers, R. Le Bras, J. Gao, and Y. Choi, “PIQA: Reasoning about physical commonsense in natural language,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 5, pp. 7432–7439, 2020. DOI: 10.1609/aaai.v34i05.6239

- [26] A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman, “GLUE: A multi-task benchmark and analysis platform for natural language understanding,” in *International Conference on Learning Representations*, 2019.
- [27] S. Biderman, H. Schoelkopf, L. Sutawika, L. Gao, J. Tow, B. Abbasi, et al., “Lessons from the trenches on reproducible evaluation of language models,” *arXiv preprint arXiv:2405.14782*, 2024.